Emergent Communication between Heterogeneous Visual Agents through Decentralized Learning

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Abstract

Symbols are shared, but perception is private. We study emergent communication between heterogeneous visual agents through decentralized learning, asking what visual information can become shareable when agents have different visual representations. Instead of optimizing messages through a shared external communicative objective, our agents exchange only discrete token sequences and update their own models using local perceptual evidence. This setting focuses on an underexplored aspect of emergent communication, examining whether common symbols can arise without shared perceptual access, and how the similarity between private visual spaces constrains the content and symmetry of the resulting language. We instantiate this setting in the Metropolis-Hastings Captioning Game (MHCG), where two agents collaboratively form shared captions by exchanging proposed token sequences that a listener accepts or rejects using an MH-style criterion evaluated against its own visual features. We compare three pairings of frozen visual encoders, with agents starting from randomly initialized text modules. Experiments on MS-COCO show that MHCG produces visually informative shared token sequences that outperform a no-communication baseline in cross-agent alignment, visual-feature prediction, and image-text retrieval; all cross-agent metrics decline as encoder mismatch increases. Moderate encoder heterogeneity reduces the number of shared sequences while preserving per-sequence visual specificity, whereas stronger encoder heterogeneity yields fewer, coarser, and more asymmetric sequences. Ablations show that listener-side MH acceptance is critical for avoiding degenerate token formation. These results suggest that shared symbols can arise from local perceptual evaluation alone, with visual representational similarity across encoders shaping both the content and symmetry of the resulting language.

1 Introduction

Communication requires agents to coordinate public symbols, but each agent grounds those symbols in a private perceptual representation that others cannot directly observe[1]. Such visual representations may differ across architectures, training histories, embodiments, or sensory systems[33]. Public symbols therefore act as an interface between private visual spaces, raising the question of which aspects of visual representations can be preserved in symbols that agents come to share. More specifically, when agents' visual representations differ, what visual information remains shareable through public symbols, and do the resulting symbols align with both agents' representational structures or privilege one agent's visual space [29]?

Emergent communication (EC) is the computational framework most directly suited to studying such questions, examining how language-like communication systems arise between interacting agents [13, 23]. The dominant paradigm in EC is referential signaling, in which a speaker's message must enable a listener to identify a target image, and emergent symbols are shaped by receiver-success

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rewards or gradients through relaxed message channels [14, 7, 21]. Within this paradigm, much work has examined the *form* of emergent symbols (compositionality, object grounding, sequence structure), and a more recent line varies visual backbones or sensory modalities to study representational heterogeneity [18, 24]. However, the shared external task signal that anchors this paradigm couples agents through a common communicative objective, which limits the leverage on our question because emergent symbols may become useful by being optimized for the shared signal rather than by being locally grounded in each agent's private perceptual evidence. This motivates a decentralized setting in which agents are coupled only through their exchanged messages and update from their own perceptual evidence [2, 27, 31, 28].

Collective Predictive Coding (CPC) formalizes such a decentralized setting, modeling symbols as shared latent variables that each agent infers from its own private observations [27, 31]. CPC has recently been derived as a specific extension of free-energy minimization to symbol emergence [30], consistent with extensions of the free-energy principle [4] to multi-agent settings through active inference accounts of dyadic communication [5] and federated inference for belief sharing under a common world model [6]. Decentralized inference in the CPC framework can be approximated by language games based on Metropolis–Hastings (MH) sampling [28, 11, 10, 9]. In these games agents alternate speaker and listener roles and the listener accepts or rejects proposed symbols through a likelihood ratio, approximating posterior inference over the shared latent variable [28]. Motivated by this line of MH language-game research, Matsui et al. [19] introduced the Metropolis–Hastings Captioning Game (MHCG) for vision–language model (VLM) agents on natural images, using ProbVLM [34] to estimate the likelihood used in the MH acceptance step.

We use MHCG to study how heterogeneity between agents' visual representations shapes the symbols they come to share. Our setting differs from Matsui et al. [19] in using randomly initialized text modules and heterogeneous frozen visual encoders. Thus, the exchanged token sequences can become communicative only through interaction, rather than by inheriting semantics from pretrained language modules (Section 2). We implement this setting with two BLIP-style multimodal agents [15] that observe independently augmented views of the same image and exchange token sequences in the captioning game (Figure 1). We evaluate our system on MS-COCO [16] under three visual-encoder conditions, each defined by a pair of frozen encoders: one homogeneous condition and two heterogeneous conditions. To connect the emergent token sequences back to the representational question above, we measure whether the two agents produce similar token-sequence structures and whether those structures align with visual representations using representational similarity analysis (RSA) [12, 26]. We additionally evaluate the visual information carried by the token sequences through cross-agent visual-feature prediction and image-text retrieval. We further assess whether the MH likelihood-ratio filter provides a useful acceptance mechanism in this VLM setting by comparing it with listener-side discriminative matching, rate-matched random acceptance, and the absence of acceptance.

Contributions.

1. We introduce a decentralized VLM language-game setting for asking what visual representational structure becomes shareable between agents with heterogeneous visual encoders, without shared communicative task rewards, cross-agent task losses, cross-agent gradients, or parameter sharing.

2. We show that MHCG produces shared token sequences that preserve visual information across agents, outperforming a no-communication baseline in inter-agent text-text alignment, text-vision alignment, visual-feature prediction, and image–text retrieval.

3. We characterize how encoder heterogeneity changes the resulting shared symbols: all cross-agent metrics decrease with encoder mismatch; moderate mismatch reduces the number of shared sequences while preserving per-sequence visual specificity; and strong mismatch produces fewer and coarser sequences with a directional bias toward one agent's representational structure

4. We show that listener-side MH likelihood-ratio filtering is important in this VLM instantiation, preventing degenerate token formation and supporting visual-information transfer compared with discriminative, random, all-accept, and no-communication controls.

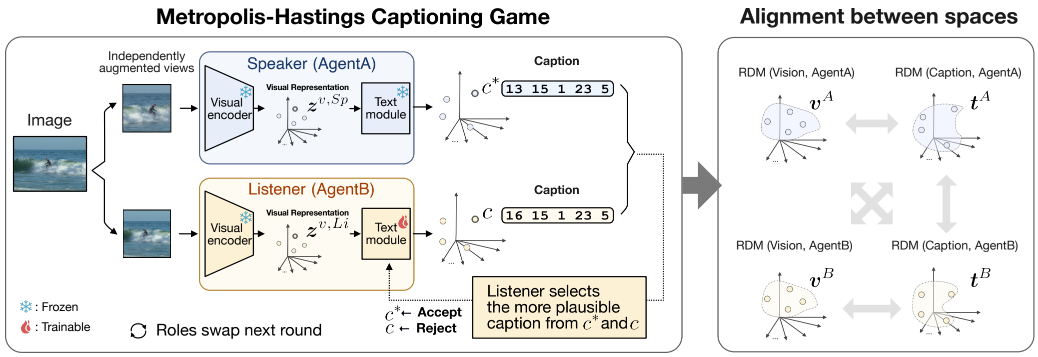

Figure 1: Metropolis–Hastings Captioning Game for heterogeneous visual agents. Two agents observe independently augmented views of the same image through private frozen visual encoders and exchange only discrete token sequences. The speaker proposes a caption c^*, and the listener compares it with its own caption c, accepting the proposal or retaining its own caption as the learning target according to listener-side visual evidence. The speaker and listener roles swap in the next round. We construct representational dissimilarity matrices (RDMs) of pairwise distances within visual representations and caption, and correlate them through the representational similarity analysis (RSA). This operationalizes the central question of whether public token sequences preserve information from both private visual spaces, or whether encoder mismatch makes the emergent symbols align more strongly with one agent's visual geometry.

2 Related Work

Heterogeneous agents in emergent communication. Most EC studies of agent diversity or perceptual heterogeneity still ground communication in a shared task signal, leaving open how shared symbols form when agents rely only on private perceptual evidence. Some EC studies introduce diversity at the population or training level: Rita et al. [25] studied how population heterogeneity shapes emergent languages, and Michel et al. [20] found that limiting co-adaptation across sender–receiver pairs improves compositionality. These works vary population composition or training dynamics while keeping perception identical across agents. More directly related to our setting, a separate line of work has begun to vary the perceptual systems themselves. Mahaut et al. [18] studied referential communication in communities of heterogeneous pre-trained visual networks, showing that such networks can learn a shared self-supervised protocol that captures high-level semantic features. Pitzer and Mihai [24] studied EC in agents with different sensory modalities. Unlike these referential-game settings, our agents are not optimized for receiver success under a shared external task signal. They update only from evidence evaluated within their own visual representations. We do not include a referential-game baseline, because standard referential games optimize messages using receiver success rewards or task losses, introducing the shared external signal that our decentralized setting is designed to exclude.

3 Methods

Architecture Each agent consists of a frozen visual encoder and randomly initialized text modules connected to a shared probabilistic embedding space through ProbVLM. Where Matsui et al. [19] represent each agent's internal state as a single latent variable, we decompose it into modality-specific features $(z^{n,v}, z^{n,t})$ (Appendix A). We realize this structure with a BLIP-style multimodal architecture [15] (single-agent internals illustrated in Figure 4 of Appendix B; see also Algorithm 1). Each agent $n \in \{A, B\}$ has a frozen pretrained ViT-based visual encoder $\mathrm{Enc}^{n,v}$ [3], whose specific instantiation depends on the encoder condition defined in Section 4. The visual encoder produces visual features $z^{n,v} = \mathrm{Enc}^{n,v}(o)$ from an input image o. Each agent also has a text encoder/decoder that uses the BERT-Tiny configuration from the 24 smaller BERT models released with the google-research BERT repository [32], but all text-module weights are initialized from scratch; no pretrained BERT checkpoint is loaded. This module is parameterized by θ^n and shares Feed-Forward and Self-Attention layers between encoder and decoder modes [15]: the encoder maps token sequences

c to text-side features $z^{n,t}$, and the decoder Dec^n generates token sequences conditioned on $z^{n,v}$. An aggregation function ψ (by default, first-token selection) followed by a linear projection maps encoder outputs into a shared latent space: $\boldsymbol{\mu}^{n,m} = \mathrm{Proj}^{n,m}(\psi(z^{n,m}))$ for modality $m \in \{v, t\}$. A ProbVLM adapter [34] (parameterized by $\phi^{n,m}$) then maps each projected embedding to diagonal density parameters:

$$\hat{\lambda}^{n,m} = \mathrm{ProbVLM}(\boldsymbol{\mu}^{n,m}; \phi^{n,m}), \quad m \in \{v, t\}. \tag{1}$$

In the implementation, $\hat{\lambda}^{n,m}$ consists of a location and scale for a diagonal generalized Gaussian distribution with shape parameter fixed to $\beta = 2$, i.e., a diagonal Gaussian up to the ProbVLM scale parameterization. During MH acceptance, the listener samples a stochastic visual embedding $\hat{h}^{\mathrm{Li},v}$ from its vision-side density, while the text-side density parameters $\hat{\lambda}^{\mathrm{Li},t}(c)$ define the likelihood assigned to a candidate caption (Eq. 3).

Metropolis–Hastings Captioning Game Following Matsui et al. [19], we use a one-sample, one-step MH-style acceptance filter motivated by decentralized Bayesian inference over the intractable posterior $p(c \mid o^A, o^B)$. The training procedure alternates between caption updates (MH acceptance, used as an E-step-like update) and parameter updates (a surrogate M-step); the correspondence between the BLIP losses and this update is discussed in Appendix A.

The listener accepts or rejects the speaker's proposed token sequence through a likelihood ratio evaluated against its own visual features. The agents alternate speaker (Sp) and listener (Li) roles; the speaker encodes its observation and samples a candidate token sequence:

$$c^* \sim q(c^* \mid z^{\mathrm{Sp},v}; \theta_{\mathrm{Sp}}). \tag{2}$$

The listener also generates a current caption c from its own decoder for the same image. It then decides whether to accept the speaker proposal c^* or retain this listener-generated caption. The target posterior factorizes as $\pi(c) := p(c \mid z^{\mathrm{Sp},v}, z^{\mathrm{Li},v}) \propto p(z^{\mathrm{Li},v} \mid c)\, p(c \mid z^{\mathrm{Sp},v})$ by conditional independence of the agents given c, which follows from the CPC generative model [27, 19], where $p(c \mid z^{\mathrm{Sp},v})$ is the speaker's posterior over captions and $p(z^{\mathrm{Li},v} \mid c)$ is the listener's likelihood of its visual features under caption c, evaluated through the ProbVLM density (Eq. 1). The proposal distribution is the speaker's conditional generative distribution $q(c^* \mid z^{\mathrm{Sp},v})$. Operationally, we use a one-step MH-style independence update in which the listener compares the speaker proposal against its own current caption. Under the *self-consistency approximation* $p(c \mid z^{\mathrm{Sp},v}) \approx q(c \mid z^{\mathrm{Sp},v})$ [19], the speaker-dependent terms in the standard MH ratio cancel, and the acceptance probability simplifies to a likelihood ratio from the listener's perspective:

$$r \;=\; \min\left(1,\; \frac{p(\hat{h}^{\mathrm{Li},v} \mid \hat{\lambda}^{\mathrm{Li},t}(c^*))}{p(\hat{h}^{\mathrm{Li},v} \mid \hat{\lambda}^{\mathrm{Li},t}(c))}\right), \tag{3}$$

where $\hat{h}^{\mathrm{Li},v}$ is a sample from the listener's visual ProbVLM distribution and $\hat{\lambda}^{\mathrm{Li},t}(c)$ denotes the listener's text-side ProbVLM density parameters predicted from caption c. Because this is not a persistent chain and the self-consistency approximation is weak at initialization, we use the ratio in Eq. 3 as an approximate, listener-side perceptual acceptance rule rather than claiming exact posterior sampling; Appendix B.1 discusses this approximation and its empirical diagnostics, where the finite-support Jensen–Shannon divergence decreases from 0.660 at epoch 0 to approximately 0.35 at epoch 40, and the Spearman correlation between decoder and encoder/ProbVLM log scores increases from -0.03 to approximately 0.77.

For each image and communication direction in an MH epoch, one speaker proposal and one listener current caption are generated, and one accept/reject decision is made. The selected caption (the proposal if accepted, otherwise the listener-generated caption) becomes the training target for the listener on that image. The opposite direction is then processed analogously.

Each agent updates its text modules using the accepted token sequences as training targets, while its visual encoder remains frozen throughout. Parameter updates are interleaved with the two ordered speaker–listener role assignments: in each MH epoch, Agent A first serves as the speaker and Agent B as the listener for one pass over the dataset; the captions selected by Agent B's MH acceptance rule are then used to update Agent B. The same procedure is then applied with roles reversed (B speaks, A listens, A is updated). The text encoder, text decoder, projection layers, and ITM head (collectively

θ_n) are updated using the selected captions as fixed targets, minimizing BLIP-style objectives [15]:

$$\mathcal{L} \;=\; \mathcal{L}_{\mathrm{ITC}} + \mathcal{L}_{\mathrm{ITM}} + \mathcal{L}_{\mathrm{LM}}, \tag{4}$$

where $\mathcal{L}_{\mathrm{ITC}}$ is image-text contrastive loss, $\mathcal{L}_{\mathrm{ITM}}$ is image-text matching loss, and $\mathcal{L}_{\mathrm{LM}}$ is language modeling loss. The ProbVLM adapter ϕ_n is updated separately using $\mathcal{L}_{\mathrm{ProbVLM}}$ [34], with all other parameters frozen. The full procedure is summarized in Algorithm 1 (Appendix B).

4 Experimental Setup

Dataset and transforms. We train and evaluate on MS-COCO [16], applying independently sampled augmentations to each agent during interaction so that the two agents observe distinct views of the same image. Training and interaction are conducted on `train2017` approximately 118K images), and all evaluations are performed on `val2017` (5K images), which is unseen during interaction. During training, the two agents observe the same underlying image through independently sampled augmentations: Random Resized Crop with scale 0.8–1.0 and Random Horizontal Flip. All reported validation metrics use a deterministic validation transform without random crop or flip.

Visual encoder conditions. We compare three visual-encoder conditions, each defined by a pair of frozen encoders: one homogeneous and two heterogeneous (Table 1). The two agents always see independently augmented views during interaction, so view-level difference is a shared premise of all conditions. The conditions use the same dataset, augmentation protocol, and interaction procedure, and differ only in the frozen visual encoders assigned to the two agents.

We instantiate the conditions using DINOv2 visual encoders [22] and an MAE visual encoder [8].

Table 1: Three visual encoder conditions.

Condition	Agent A	Agent B
HOMO	DINOv2 ViT-B/14	DINOv2 ViT-B/14
HETERO1	DINOv2 ViT-B/14	DINOv2 ViT-S/14
HETERO2	DINOv2 ViT-B/14	MAE ViT-B/16

HOMO pairs identical DINOv2 ViT-B/14 encoders and is the homogeneous reference condition: during interaction, the agents still receive independently sampled image views, but their frozen visual encoders are the same. HETERO1 pairs DINOv2 ViT-B/14 with DINOv2 ViT-S/14 as the first heterogeneous encoder condition (D^v=768 vs. D^v=384). HETERO2 pairs DINOv2 ViT-B/14 with MAE ViT-B/16 as the second heterogeneous encoder condition. We refer to HETERO1 as moderate heterogeneity and HETERO2 as stronger heterogeneity, where heterogeneity is quantified by vision–vision RSA: the Spearman correlation between the vectorized visual RDMs constructed from each frozen encoder's `[CLS]` features, $\rho_s(\boldsymbol{v}^A, \boldsymbol{v}^B)$ (Table 3).

Communication and acceptance-rule controls. We compare MHCG against a no-communication baseline (NoCom, $r \equiv 0$) and, for HETERO1, four acceptance-rule ablations (Table 2). These controls ask whether successful token-sequence formation requires the MH likelihood-ratio filter specifically, or can be explained by proposal exchange alone, matched acceptance rates, or the listener's BLIP-style discriminative image–text matching ability. Unless otherwise stated, each encoder condition is trained with the default MHCG interaction rule, in which listener-side Metropolis–Hastings acceptance determines whether a proposed token sequence is adopted. The HETERO1 controls use the same data, encoders, model architecture, and training horizon as the corresponding MHCG run, differing only in the communication or acceptance rule.

Evaluation Metrics We evaluate whether the emergent token sequences align across agents, how much visual information they transfer, and whether images assigned the same shared token sequence form compact clusters in fixed visual feature spaces. All validation metrics are measured on `val2017`; visual-feature prediction probes are trained on `train2017` and evaluated on `val2017`. Full metric definitions are provided in Appendix B.6. Throughout this section, $X\!\rightarrow\!Y$ denotes a token-sequence-source / evaluation-space pairing: Agent X produces the token sequence, and Agent Y supplies the representation space used for evaluation. We use Representational Similarity Analysis (RSA), which

Table 2: Communication and acceptance-rule controls.

Method	Acceptance rule
MHCG	Listener-side MH acceptance using the ProbVLM likelihood ratio
NoCom	$r \equiv 0$; never accepts proposed token sequences
AllAccept	$r \equiv 1$; accepts every proposal
Random accept	Matches the MHCG acceptance rate, but accepts uniformly at random
ITM-based	Uses the listener's BLIP-style ITM head score for the candidate image–token-sequence pair

measures correspondence between representational spaces by comparing their pairwise dissimilarity structures, encoded in Representational Dissimilarity Matrices (RDMs). In RSA notation, $\boldsymbol{t}^A$ and $\boldsymbol{t}^B$ denote token-sequence RDMs for Agents A and B, whereas $\boldsymbol{v}^A$ and $\boldsymbol{v}^B$ denote visual RDMs computed from each agent's frozen visual-encoder `[CLS]`-token output (the global image representation pooled from the first token of the ViT output), before the learned multimodal projection heads.

For cross-agent alignment, we report top-1 cross-modal retrieval ($K{=}10$) and inter-agent text–text RSA. Retrieval asks whether a token sequence produced by one agent can identify the corresponding image in the other agent's multimodal space, and text–text RSA measures convergence between the agents' token-sequence geometries. For visual information transfer, we use two text-to-vision measures: ΔR^2, a permutation-corrected linear prediction score from token indicators to visual PCs, and vision–text RSA, the Spearman correlation between token-sequence and visual RDMs. Within-agent pairings ($A{\to}A$, $B{\to}B$) measure how much visual information each agent's token sequence carries about its own visual space, whereas cross-agent pairings ($A{\to}B$, $B{\to}A$) measure transfer to the other agent's visual space. We measure the visual specificity of each shared token sequence's image set using the global-normalized visual radius in fixed CLS measurement spaces, distinct from the agent-specific spaces used during training. A token sequence is shared within a seed if at least 10 validation images receive that sequence from both agents. Visual specificity captures the narrowness of each shared concept's image set and is distinct from the total amount of visual information carried by the entire code.

5 Results and Discussions

Encoder mismatch reduces cross-agent communicability. In every encoder condition, MHCG outperformed all cross-agent metrics over NoCom, with performance decreasing as encoder mismatch increased (Table 3). The absolute performance decreased as the visual encoders became less similar, with HOMO generally highest and HETERO1 below it, but even HETERO2 remained above the no-communication baseline. Thus, emergent token sequences encode visual information that transfers across agents. Direction-wise and positional details are in Appendix Table 8, 13, 14.

Moderate heterogeneity yields fewer shared sequences but no systematic reduction in per-sequence visual specificity. We evaluate per-sequence visual concentration in fixed CLS measurement spaces, distinct from the agent-specific spaces that shaped their formation, to avoid artificially inflating specificity scores. We computed the global-normalized visual radius of each shared token sequence in three fixed CLS spaces: DINOv2 ViT-B, DINOv2 ViT-S, and MAE ViT-B. HETERO1 showed numerically smaller mean and median radii than HOMO across all three measurement spaces (Table 4), but formed 39% fewer shared sequences. Bootstrap CIs of the per-seed radius difference overlapped zero in two of three seeds, so the per-sequence difference should be treated as suggestive rather than statistically established at $n{=}3$ seeds. Accounting for shared-sequence count, HOMO has greater system-level vocabulary coverage, consistent with its higher ΔR^2 reported above.

Figure 2 shows that all three conditions produced visually coherent shared token sequences corresponding to recognisable visual categories. Comparing HOMO and HETERO1, the C1–C3 image sets were largely aligned, with boundary shifts rather than wholesale replacement (Appendix Figure 7). HETERO2 yields far fewer shared token sequences, each covering a larger and more object-diverse set of images (Table 9), consistent with the markedly larger visual radii reported above.

Representational bias toward one encoder's visual space grows with encoder mismatch. Shared token sequences do not reflect both agents' visual geometries equally: Partial-RSA bias Δ_X (Table 5)

Table 3: Communication summary at epoch 40 on COCO `val2017`. Rows report seed means ± SD. Cross-agent columns average the $A \to B$ and $B \to A$ directions; retrieval is top-1 accuracy with K=10. Underlines mark MHCG values above the corresponding NoCom row. Full direction-wise means and standard deviations are in Appendix Table 8.

Condition	Method	$\rho_s(\boldsymbol{v}^A, \boldsymbol{v}^B)\uparrow$	$\rho_s(\boldsymbol{t}^A, \boldsymbol{t}^B)\uparrow$	Cross-agent $\rho_s(\boldsymbol{t}, \boldsymbol{v})\uparrow$	Cross-agent $\Delta R^2 \uparrow$	Cross-agent retrieval i2t (%) ↑	Cross-agent retrieval t2i (%) ↑
HOMO	MHCG	1.000	<u>0.937</u>±0.021	<u>0.162</u>±0.054	<u>0.312</u>±0.035	<u>92.3</u>±1.4	<u>75.5</u>±4.6
	NoCom		0.044±0.057	0.079±0.019	0.050±0.017	23.0±6.1	9.6±0.7
HETERO1	MHCG	0.417	<u>0.872</u>±0.032	<u>0.127</u>±0.039	<u>0.286</u>±0.004	<u>87.6</u>±2.6	<u>62.0</u>±7.1
	NoCom		0.042±0.012	0.053±0.025	0.039±0.015	27.0±4.3	9.8±0.5
HETERO2	MHCG	0.074	<u>0.673</u>±0.086	<u>0.088</u>±0.047	<u>0.119</u>±0.073	<u>59.0</u>±7.1	<u>35.5</u>±12.0
	NoCom		0.015±0.005	0.037±0.013	0.025±0.016	22.4±6.0	10.0±0.2

Table 4: Global-normalized visual radius in fixed CLS measurement spaces. Seed means ± SD over three seeds; lower = more visually specific; underlines mark row minima.

Measurement space	Condition	Mean radius ↓	Median radius ↓
DINOv2 ViT-B	HOMO	0.525±0.016	0.556±0.011
	HETERO1	<u>0.505</u>±0.056	<u>0.499</u>±0.144
	HETERO2	0.654±0.060	0.804±0.092
DINOv2 ViT-S	HOMO	0.530±0.011	0.539±0.016
	HETERO1	<u>0.499</u>±0.042	<u>0.492</u>±0.128
	HETERO2	0.644±0.067	0.758±0.142
MAE ViT-B	HOMO	0.737±0.052	0.699±0.026
	HETERO1	<u>0.733</u>±0.010	<u>0.691</u>±0.008
	HETERO2	0.761±0.140	0.782±0.215

showed $\Delta_A > 0$ and $\Delta_B < 0$ under both heterogeneous conditions, indicating a consistent directional tendency toward Agent A's (DINOv2 ViT-B/14) visual geometry; however, the magnitude and behavioral consequences of this bias grow with encoder mismatch. In HETERO1, where Agent A is the higher-capacity encoder (ViT-B vs. ViT-S) and their CLS-derived visual RDMs remain moderately aligned ($\rho_s(\boldsymbol{v}^A, \boldsymbol{v}^B)$=0.417), this bias ($\Delta_A$=0.031±0.029) does not produce retrieval asymmetry ($A \to B$=87.6±2.6%, $B \to A$=87.5±2.6% i2t). Under HETERO2, where the CLS-derived visual RDMs are far more dissimilar ($\rho_s(\boldsymbol{v}^A, \boldsymbol{v}^B)$=0.074), the bias is larger (Δ_A=0.077±0.050) and translates into a clear retrieval asymmetry: Agent B's token sequences transfer more readily to Agent A's evaluation space than vice versa ($A \to B$=47.7±13.8%, $B \to A$=70.3±3.7% i2t; Appendix Table 8).

Listener-side MH acceptance prevents token-sequence collapse and supports visual-information transfer. Listener-side MH acceptance is critical to prevent collapse and support visual-information transfer in the tested HETERO1 condition. We ablated the acceptance rule in the HETERO1 configuration, a moderately heterogeneous setting in which the default system performs well. Table 6 summarizes these controls using the same summary metrics as Table 3; full direction-wise values and seed standard deviations are reported in Appendix Table 8. MHCG was the strongest method in every column. The listener-side MH acceptance filter both prevented degenerate token-sequence formation and enabled the token sequences to carry visual information across agents.

The individual controls revealed distinct failure modes. AllAccept produced high inter-agent text–text RSA, but this agreement was degenerate: compared with MHCG, it sharply reduced the number of unique sequences (3.5±2.5 vs. 90.8±34.0) and cross-agent retrieval (i2t/t2i: 52.8/15.8% vs. 87.6/62.0%). Random accept separated the acceptance rate from the acceptance criterion: it accepted proposals at a similar rate to MHCG but remained near chance in text-to-image retrieval and carried little visual information. ITM-based accept improved over Random accept, but remained below MHCG across visual prediction, cross-agent vision–text RSA, and retrieval. Together, these controls show that useful token-sequence formation requires listener-side perceptual filtering, not merely proposal exchange, high acceptance, or discriminative image–text matching.

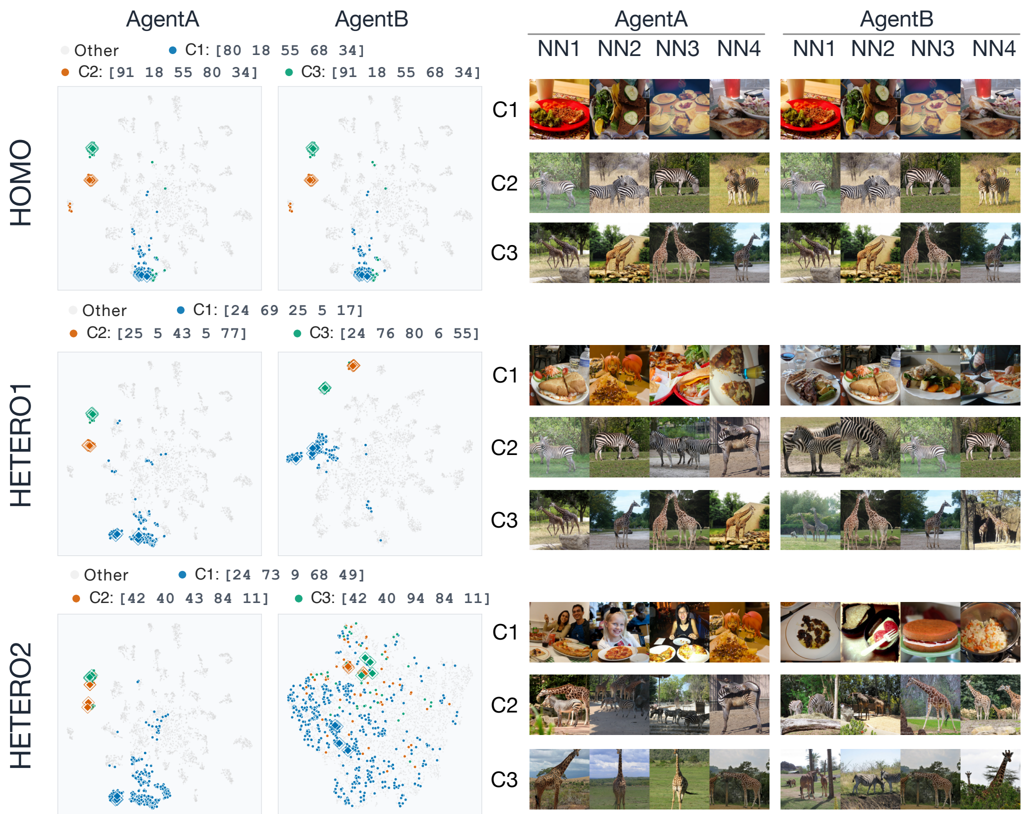

Figure 2: Matched image-set concepts across encoder conditions. C1–C3 are token-sequence triplets matched by image-set overlap after requiring both agents to assign the identical sequence to the same images. The t-SNE panels show each agent's visual space colored by the matched concepts, and the image strips show nearest neighbors around each concept centroid. Additional examples are in Appendix C.4.

Table 5: Private-structure bias of emergent token sequences. Bias is computed as $\Delta_X = \rho_{X,X|Y} - \rho_{X,Y|X}$; positive values indicate stronger dependence on Agent X's own visual geometry.

Condition	Δ_A	Δ_B
HETERO1	0.031±0.029	−0.024±0.031
HETERO2	0.077±0.050	−0.037±0.021

Table 6: HETERO1 acceptance-rule ablations at epoch 40. Rows report seed means ± SD over available seeds. Cross-agent columns average $A \to B$ and $B \to A$, and the fixed vision–vision RSA (0.417) is omitted. Underlines mark column-wise maxima.

Method	$\rho_s(\boldsymbol{t}^A, \boldsymbol{t}^B) \uparrow$	Cross-agent $\rho_s(\boldsymbol{t}, \boldsymbol{v}) \uparrow$	Cross-agent $\Delta R^2 \uparrow$	Cross-agent retrieval i2t (%) ↑	Cross-agent retrieval t2i (%) ↑
MHCG	$\underline{0.872}$±0.032	$\underline{0.127}$±0.039	$\underline{0.286}$±0.004	$\underline{87.6}$±2.6	$\underline{62.0}$±7.1
NoCom	0.042±0.012	0.053±0.025	0.039±0.015	27.0±4.3	9.8±0.5
AllAccept	0.774±0.236	0.094±0.021	0.037±0.035	52.8±3.1	15.8±5.0
Random accept	0.011±0.023	0.016±0.017	0.009±0.004	47.8±5.2	10.0±0.0
ITM-based	0.718±0.166	0.100±0.043	0.099±0.030	67.8±9.8	38.3±8.9

6 Conclusions, Limitations and Future Work

We asked which aspects of visual representational structure can be preserved in shared symbols formed by agents with different private visual representations, and whether those symbols align with both agents' representational structures or reflect one agent's structure more strongly. We showed that MHCG produces shared symbols, instantiated as discrete token sequences, that carry visual information across agents and outperform a no-communication baseline in cross-agent text–text alignment, text–vision alignment, visual-feature prediction, and image–text retrieval. Across the homogeneous and heterogeneous encoder conditions, all cross-agent metrics decreased with increasing encoder mismatch. Moderate heterogeneity reduced the number of shared sequences while preserving per-sequence visual specificity, whereas strong heterogeneity yielded fewer and coarser shared token sequences. Partial-RSA analysis further suggested a directional tendency toward one agent's visual structure: Δ_A was positive and Δ_B negative on average in both heterogeneous conditions, but the effect was small under moderate mismatch (Δ_A=0.031±0.029 in HETERO1) and larger under strong mismatch (Δ_A=0.077±0.050 in HETERO2), where it was accompanied by a clear asymmetry in retrieval accuracy.

HETERO1 formed fewer shared sequences than HOMO, yet the per-sequence visual concentration was similar across all three fixed measurement spaces. This suggests a possible precision–coverage tradeoff, in which moderate heterogeneity may preserve the visual specificity of individual shared sequences while reducing the size of the shared vocabulary. The lower visual-feature prediction performance under moderate heterogeneity is consistent with this reduced coverage, because fewer distinct visual concepts are named and less visual information transfers across agents overall. Whether the preserved per-sequence specificity reflects genuine sharpening under heterogeneous constraints or a selection bottleneck that retains only encoder-invariant concepts remains unresolved at n=3 seeds. Future work with larger seed sets, controlled vocabulary sizes, and extended training could decouple these explanations and determine whether heterogeneous constraints can improve per-sequence concept quality without sacrificing vocabulary coverage.

Several limitations bound the scope of our conclusions, but they also point to larger questions about how perceptual and linguistic representations co-develop through interaction. A broader goal is to understand how private visual representations and shared symbolic systems are jointly constructed. Our study isolates one part of this problem by freezing the visual encoders, which leaves open what happens when perception itself adapts during communication. Future work should ask whether shared symbols reshape private perceptual spaces, not only whether fixed perceptual spaces constrain shared symbols.

A second limitation is the restricted range of empirical conditions: we tested only three encoder pairings on a single natural-image dataset. Results of HETERO2 suggests that large representational gaps can produce coarser and more asymmetric shared token sequences, yet additional encoder families, sensory modalities, datasets, and larger agent populations are needed to determine whether this pattern reflects a general relation between representational heterogeneity and symbol sharing. Third, the vocabulary ($|\mathcal{V}| = 100$) and sequence length ($L = 5$) define a small token-sequence space. Larger vocabularies or longer sequences may support more fine-grained concepts while making decentralized coordination harder.

Finally, our MHCG update is a one-step MH-style approximation rather than a persistent Markov chain, and the cancellation of speaker-dependent terms relies on a self-consistency approximation between the decoder proposal distribution and the speaker-side posterior. Appendix B.1 shows that this approximation is weakest at random initialization and may affect the early dynamics of token-sequence formation. Future work should investigate better calibrated acceptance ratios, asymmetric learning rates, auxiliary alignment objectives, and more scalable variants of decentralized inference under strong perceptual heterogeneity.

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A Correspondence between BLIP Training Losses and the Approximate M-step

We relate the MHCG training procedure to a Monte Carlo EM-like update for the CPC generative model, and clarify which part of the BLIP training objective corresponds to the parameter update after MH caption sampling. The correspondence is approximate: the MH step supplies a sampled caption state, while the neural update optimizes a conditional captioning surrogate rather than the exact complete-data likelihood of the full generative model.

The CPC generative model assumes a shared latent token sequence $c \in \mathcal{V}^L$ that generates each agent's internal representations:

$$p(c, Z^A, Z^B, o^A, o^B;\, \theta, \phi) = p(c) \prod_{n \in \{A,B\}} p(Z^n \mid c;\, \theta_n, \phi_n)\, p(o^n \mid Z^n), \tag{5}$$

where $\theta = (\theta_A, \theta_B)$ denotes the BLIP-style VLM parameters and $\phi = (\phi_A, \phi_B)$ denotes the ProbVLM parameters.[1] Matsui et al. [19] represent Z^n as a single latent variable. We decompose it into three layers: $Z^n = \{z^{n,t},\, h^n,\, z^{n,v}\}$, where $z^{n,t}$ are text features produced by the text encoder, $z^{n,v}$ are visual features produced by the frozen visual encoder, and h^n is a probabilistic embedding in a shared latent space produced by ProbVLM. The conditional factorizes as:

$$p(Z^n \mid c;\, \theta_n, \phi_n) = p(z^{n,t} \mid c;\, \theta_n)\, p(h^n \mid z^{n,t};\, \theta_n, \phi_n)\, p(z^{n,v} \mid h^n;\, \theta_n, \phi_n). \tag{6}$$

This decomposition allows each modality to maintain its own feature space while h^n provides a probabilistic bridge between modalities (Section 3).

A.1 Approximate M-step Derivation

In an ideal MCEM treatment, the E-step would draw caption samples from the posterior $p(c \mid O;\, \theta^{\text{old}}, \phi^{\text{old}})$ and the M-step would maximize the expected complete-data log-likelihood. In MHCG, the MH communication step plays the role of an approximate E-step: for each image it returns a selected caption state $c^{(s)}$. With one Monte Carlo sample, the exact per-agent M-step would involve the marginal likelihood

$$\max_{\theta_n, \phi_n} \log p(o^n, c^{(s)};\, \theta_n, \phi_n) = \max_{\theta_n, \phi_n} \log p(o^n \mid c^{(s)};\, \theta_n, \phi_n) + \text{const.}, \tag{7}$$

where the caption prior is independent of the agent parameters.

In our three-layer generative model (Eq. 6), the conditional $p(o^n \mid c^{(s)};\, \theta_n, \phi_n)$ involves marginalizing over all intermediate latent variables:

$$p(o^n \mid c^{(s)};\, \theta_n, \phi_n) = \iiint p(z^{n,t} \mid c^{(s)};\, \theta_n)\, p(h^n \mid z^{n,t};\, \theta_n, \phi_n)\, p(z^{n,v} \mid h^n;\, \theta_n, \phi_n)\, p(o^n \mid z^{n,v})\, \mathrm{d}z^{n,t}\, \mathrm{d}h^n\, \mathrm{d}z^{n,v}. \tag{8}$$

This integral is intractable and is not the objective optimized by the implemented BLIP update. Instead, because the visual encoder is frozen, each observation can be represented by a fixed feature $z^{n,v} = \mathrm{Enc}^{n,v}(o^n)$, and the VLM update trains the decoder to predict the selected caption from this fixed visual feature. We therefore use the following conditional surrogate objective for the text module:

$$\theta_n^{\text{new}} \approx \arg\max_{\theta_n} \log q_{\theta_n}(c^{(s)} \mid z^{n,v}), \tag{9}$$

where q_{θ_n} denotes the decoder distribution implemented by the BLIP-style text decoder. This is a generalized or surrogate M-step: it increases the conditional likelihood of the sampled caption under the agent's own visual representation, but it is not claimed to be the exact maximizer of Eq. 7.

Under the autoregressive factorization of the text decoder,

$$\log q_{\theta_n}(c^{(s)} \mid z^{n,v}) = \sum_{l=1}^{L} \log q_{\theta_n}(c_l^{(s)} \mid c_{<l}^{(s)},\, z^{n,v}) = -\mathcal{L}_{\text{LM}}. \tag{10}$$

[1] Following Matsui et al. [19], we do not specify the prior $p(c)$ concretely, as it is not used during training or inference.

Therefore, minimizing the language modeling loss $\mathcal{L}_{\mathrm{LM}}$ with the selected caption $c^{(s)}$ as target exactly optimizes the conditional captioning surrogate in Eq. 9. The three-layer latent structure (Eq. 6) still matters for MH acceptance, because the listener evaluates captions through the ProbVLM density over the shared latent space. It is not explicitly marginalized in the BLIP decoder update. The ProbVLM parameters ϕ_n are updated separately with $\mathcal{L}_{\mathrm{ProbVLM}}$ while θ_n is frozen, so their update is best viewed as an additional approximation that calibrates the density model used in the next MH step.

A.2 Role of Auxiliary Losses

The image-text contrastive loss $\mathcal{L}_{\mathrm{ITC}}$ and the image-text matching loss $\mathcal{L}_{\mathrm{ITM}}$ are not part of the conditional surrogate in Eq. 9. They are auxiliary objectives that maintain the representations on which subsequent MH acceptance decisions depend:

$\mathcal{L}_{\mathrm{ITC}}$ **(Image-Text Contrastive).** The contrastive loss operates on the projected embeddings $\boldsymbol{\mu}^v$ and $\boldsymbol{\mu}^t$ in the shared latent space—the space from which the probabilistic embedding h^n (Eq. 6) is derived via ProbVLM. Ignoring the momentum queue and soft targets used in the BLIP implementation, this contrastive form has the standard InfoNCE-style interpretation as a lower bound on mutual information [35]:

$$I(\boldsymbol{\mu}^v; \boldsymbol{\mu}^t) \geq \log K - \mathcal{L}_{\mathrm{ITC}}, \tag{11}$$

where K is the number of negative samples (including the memory queue). In practice, minimizing $\mathcal{L}_{\mathrm{ITC}}$ encourages a shared embedding geometry in which matched image-text pairs are close and mismatched pairs are distant. This is critical for the ProbVLM density estimates (Eq. 1) used in the MH acceptance ratio (Eq. 3): if the shared embedding space of h^n degrades, the density estimates become unreliable and the MH step no longer provides useful caption samples.

$\mathcal{L}_{\mathrm{ITM}}$ **(Image-Text Matching).** The matching loss trains a binary classifier on the fused cross-modal representation to predict whether an image-caption pair is matched. This objective encourages the cross-modal encoder to preserve discriminative information about image-caption compatibility. Without $\mathcal{L}_{\mathrm{ITM}}$, the h^n representation may lose its discriminative capacity, degrading the agent's ability to evaluate the consistency of proposed captions in the subsequent MH step.

A.3 Summary

The VLM training objective $\mathcal{L} = \mathcal{L}_{\mathrm{ITC}} + \mathcal{L}_{\mathrm{ITM}} + \mathcal{L}_{\mathrm{LM}}$ can therefore be decomposed as follows:

Loss	Role	Function
$\mathcal{L}_{\mathrm{LM}}$	Surrogate M-step	Maximize $\log q_{\theta_n}(c^{(s)} \mid z^{n,v})$
$\mathcal{L}_{\mathrm{ITC}}$	MH support	Maintain h^n embedding quality for ProbVLM
$\mathcal{L}_{\mathrm{ITM}}$	MH support	Maintain h^n discriminative capacity for MH acceptance

By alternating MHCG communication and parameter updates with $\mathcal{L}$, the agents perform an MCEM-like procedure: the MH step updates the sampled caption state, while the neural update fits each agent's conditional caption model and maintains the representation and density estimates needed for future MH rounds.

B Experiment Details

B.1 Self-Consistency Approximation Quality

The MH acceptance ratio (Eq. 3) relies on the self-consistency approximation $p(c \mid z^{\mathrm{Sp},v}) \approx q(c \mid z^{\mathrm{Sp},v})$ (Section 3). At random initialization this approximation is loose, because the shared encoder–decoder parameters have not yet learned a consistent mapping between captions and visual features. As training proceeds, the LM, ITC, and ITM objectives encourage the encoder and decoder pathways to become more compatible. We therefore evaluate the approximation as an empirical diagnostic rather than assuming it holds exactly.

The full token space has size $|\mathcal{V}|^L$, so exact normalization is infeasible. We instead compare the two sides of the approximation on a finite candidate set. For each validation image o_i, agent n, and checkpoint epoch, we construct a candidate set $\mathcal{C}_i^n$ containing the agent's greedy caption, stochastic decoder samples for the same image, and negative captions sampled from the candidate pools of other validation images. All duplicate token sequences are removed. This produces a local finite support over plausible and implausible captions without requiring enumeration of all length-L sequences.

On this support, the decoder side is measured by teacher forcing. For a candidate caption $c = (c_1, \ldots, c_L)$ and visual feature $z_i^{n,v}$, we compute

$$\ell_{\text{dec}}^n(c; o_i) = \sum_{\ell=1}^{L} \log q_{\theta^n}(c_\ell \mid c_{<\ell}, z_i^{n,v}), \tag{12}$$

with padding positions ignored if present, and normalize over $\mathcal{C}_i^n$:

$$\tilde{q}_i^n(c) = \frac{\exp(\ell_{\text{dec}}^n(c; o_i))}{\sum_{c' \in \mathcal{C}_i^n} \exp(\ell_{\text{dec}}^n(c'; o_i))}. \tag{13}$$

The encoder/ProbVLM side is measured by asking how well the caption-induced text density explains the same agent's visual embedding. Let $\bar{h}_i^{n,v}$ denote the deterministic vision-side ProbVLM location for image o_i; using the location rather than a single sample reduces Monte Carlo noise in the diagnostic. For each candidate caption, we compute

$$s_{\text{enc}}^n(c; o_i) = \log p\left(\bar{h}_i^{n,v} \mid \hat{\lambda}^{n,t}(c)\right), \tag{14}$$

where $\hat{\lambda}^{n,t}(c)$ is obtained by passing c through the text encoder, text projection, and text-side ProbVLM adapter of the same agent. Normalizing these scores over the same candidate set gives the finite-support posterior proxy

$$\tilde{p}_i^n(c) = \frac{\exp(s_{\text{enc}}^n(c; o_i))}{\sum_{c' \in \mathcal{C}_i^n} \exp(s_{\text{enc}}^n(c'; o_i))}. \tag{15}$$

We summarize the agreement between $\tilde{q}_i^n$ and $\tilde{p}_i^n$ using four quantities: the Jensen–Shannon divergence between the two normalized distributions, the Spearman correlation between their unnormalized log scores over $\mathcal{C}_i^n$, top-1 agreement of their highest-scoring candidate, and the percentile rank assigned by $\tilde{p}_i^n$ to the top candidate under $\tilde{q}_i^n$. Lower Jensen–Shannon divergence, higher rank correlation, higher top-1 agreement, and lower rank percentile indicate stronger self-consistency. We compute this diagnostic for the main HETERO1 condition (`coco_len5_aug_seed0`–`seed4`) across checkpoints, because this condition is the central heterogeneous setting used in the main analysis. The diagnostic is not used during training or model selection; it only characterizes how accurately the simplified MH-style ratio approximates the corresponding finite-support posterior comparison.

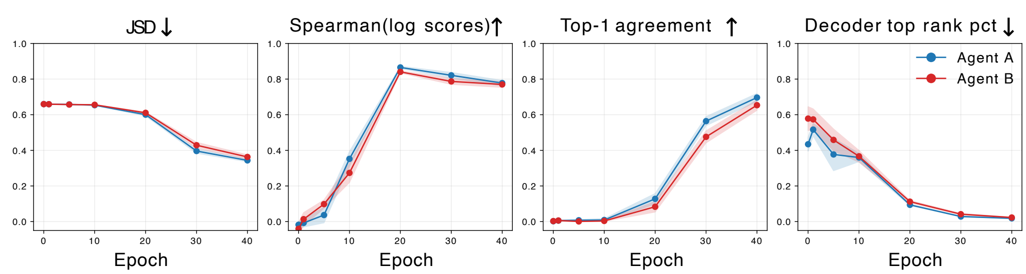


Figure 3: Self-consistency diagnostic for the main HETERO1 condition (`coco_len5_aug_seed0`–`seed4`). Curves show means over seeds, with shaded bands indicating the standard error over seeds, computed separately for Agents A and B on 512 validation images. Candidate sets contain same-image greedy and stochastic decoder captions plus captions sampled from other validation images, with duplicate token sequences removed. The encoder/ProbVLM posterior proxy uses the deterministic vision-side ProbVLM location as the visual target (`vision_mu`).

Figure 3 shows that the approximation is poor at initialization but improves during MHCG training. At epoch 0, the finite-support decoder distribution and the encoder/ProbVLM posterior proxy are nearly unrelated: the Jensen–Shannon divergence is approximately 0.660, the Spearman correlation between log scores is near zero (-0.03 when averaged over agents), and top-1 agreement is below 1%. By epoch 40, the Jensen–Shannon divergence decreases to approximately 0.35, the Spearman correlation rises to approximately 0.77, and the top-1 agreement reaches 65–70% across agents. The decoder's top-ranked caption is also assigned a much higher rank by the encoder/ProbVLM proxy: its average rank percentile falls from roughly the middle of the candidate set (0.51 at epoch 0) to the top few percent (0.02 at epoch 40), and it lies within the encoder/ProbVLM top-10% candidates in about 95% of cases.

These results support the qualitative assumption used in Eq. 3: the tied encoder–decoder architecture does not provide self-consistency at random initialization, but the BLIP training objectives make the decoder-side proposal scores and the encoder/ProbVLM-side caption scores increasingly compatible over training. Thus, the simplified acceptance rule should be interpreted as a one-step MH-style approximation whose quality improves during learning, rather than as an exact MH transition from the beginning of training.

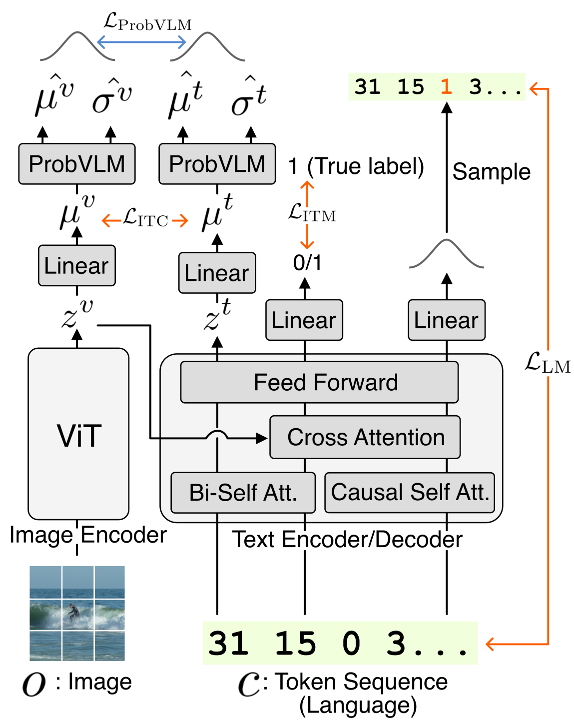


Figure 4: Single-agent BLIP-style architecture. A frozen ViT image encoder produces visual features z^v; a randomly initialized text encoder/decoder processes token sequences c. Linear projections map both modalities to a shared space ($\boldsymbol{\mu}^v$, $\boldsymbol{\mu}^t$), where ProbVLM produces density estimates used for MH acceptance. Training objectives: $\mathcal{L}_{\text{ITC}}$ (contrastive), $\mathcal{L}_{\text{ITM}}$ (matching), $\mathcal{L}_{\text{LM}}$ (language modeling), $\mathcal{L}_{\text{ProbVLM}}$ (density estimation).

B.2 Model Specifications

Table 7 summarizes the frozen visual encoders used in each condition. The DINOv2 encoders are loaded through the `timm` checkpoints corresponding to the `base_reg` and `small_reg` aliases (`vit_base_patch14_reg4_dinov2.lvd142m` and `vit_small_patch14_reg4_dinov2.lvd142m`). The MAE encoder uses the ViT-B/16 MAE checkpoint `mae_pretrain_vit_base.pth` [8]. All visual encoders operate on 224×224 images and are frozen throughout training.

Algorithm 1 Metropolis–Hastings Captioning Game (MHCG)

Require: Image dataset $\mathcal{D}$, number of epochs E_{MH}

Initialize θ^A, θ^B (text modules), ϕ^A, ϕ^B (ProbVLM)
for $i = 1, \ldots, E_{\text{MH}}$ **do**
    $\mathcal{D}^B_{\text{new}} \leftarrow \emptyset$
    for each image $o_d \in \mathcal{D}$ **do** ▷ *Agent A speaks to Agent B*
        // 1. Perception
        $z_d^{A,v} \leftarrow \text{Enc}^{A,v}(o_d), \quad z_d^{B,v} \leftarrow \text{Enc}^{B,v}(o_d)$ ▷ Different augmentations
        $\mu_d^{B,v} \leftarrow \text{Proj}^{B,v}(\psi(z_d^{B,v}))$
        $\hat{\lambda}_d^{B,v} \leftarrow \text{ProbVLM}^{B,v}(\mu_d^{B,v})$
        $\hat{h}_d^{B,v} \sim p(\hat{h} \mid \hat{\lambda}_d^{B,v})$
        // 2. Communication
        $c^* \leftarrow \text{Dec}^A(z_d^{A,v})$ ▷ Speaker proposal
        $\tilde{c}_d^B \leftarrow \text{Dec}^B(z_d^{B,v})$ ▷ Listener current caption
        $c_d^B \leftarrow \text{MH-Accept}(c^*,\, \tilde{c}_d^B,\, \hat{h}_d^{B,v})$ ▷ Eq. 3
        Add (o_d, c_d^B) to $\mathcal{D}^B_{\text{new}}$
    end for
    Update θ^B on $\mathcal{D}^B_{\text{new}}$ using $\mathcal{L}_{\text{ITC}} + \mathcal{L}_{\text{ITM}} + \mathcal{L}_{\text{LM}}$
    Update ϕ^B on $\mathcal{D}^B_{\text{new}}$ using $\mathcal{L}_{\text{ProbVLM}}$ (with θ^B frozen)
    Repeat the communication and update steps with A $\leftrightarrow$ B
end for

Table 7: Model specifications for the three visual encoder conditions.

Condition	Agent A	Agent B	Feature dimension
HOMO	DINOv2 ViT-B/14 reg	DINOv2 ViT-B/14 reg	768/768
HETERO1	DINOv2 ViT-B/14 reg	DINOv2 ViT-S/14 reg	768/384
HETERO2	DINOv2 ViT-B/14 reg	MAE ViT-B/16	768/768

The text encoder/decoder uses the BERT-Tiny configuration (L=2, H=128, A=2) from the 24 smaller BERT models released with the google-research BERT repository [32]: hidden dimension $D^t = 128$, FFN intermediate dimension 512, 2 hidden layers, 2 attention heads, max position embeddings 16. The projection dimension is $D^{\text{proj}} = 128$. The vocabulary size is $|\mathcal{V}| = 100$ and the token-sequence length is $L = 5$. The ProbVLM adapter models a diagonal generalized Gaussian density with fixed shape parameter $\beta = 2$ in the shared latent space. Visual encoders are frozen throughout training; the VLM-side trainable components are the text module, projection layers, and matching head, and the ProbVLM adapters are trained separately. All text-module weights are initialized from scratch separately for each run seed; no pretrained BERT checkpoint is used. In each communication direction, the speaker proposal and the listener current caption are both generated by the agents' decoders for that epoch; no persistent caption table is carried across epochs. Each image receives one MH-style accept/reject decision per communication direction per epoch.

B.3 Training and Decoding Details

Decoding strategy. During MHCG interaction (training), the speaker generates proposals using stochastic decoding from the text decoder with nucleus sampling ($p = 0.9$), top-k filtering ($k = 50$), and temperature annealing. This stochasticity allows the one-step accept/reject update to explore the token-sequence state space effectively. During evaluation, we switch to deterministic greedy decoding to assess the most representative token-sequence structure formed by the agents.

Training. For the VLM updates, we use AdamW [17] with $(\beta_1, \beta_2) = (0.9, 0.999)$ and weight decay 0.05. The VLM learning rate follows a global schedule over the configured MHCG horizon: 10% linear warm-up to a peak learning rate 5×10^{-5}, followed by cosine annealing to 1×10^{-6}; the epoch-40 update therefore uses a learning rate of approximately 3.8×10^{-5}. For the ProbVLM adapters, we use Adam with weight decay 1×10^{-4} and a separate per-update cosine annealing schedule (initial learning rate 1×10^{-4}, decaying to 1×10^{-5}). For each MHCG iteration, the VLM is trained for one epoch and the ProbVLM adapter for three epochs. The configured MHCG horizon is $E_{\mathrm{MH}} = 100$, with a batch size of 256 per GPU; the runs reported here are stopped after 40 MHCG epochs using `-stop_at_epoch 40` and evaluated at the epoch-40 checkpoint. Following the original BLIP implementation, we employ a momentum model and memory queues (size 131,072) for contrastive learning. Optimizer states are re-initialized for each MH update, while the VLM global step counter is carried across MH epochs so that the learning-rate and soft-label schedules remain defined over the 100-epoch horizon. The soft-label weight is linearly ramped from 0.0 to 0.4 during the first 60% of this horizon (60 MH epochs), reaching approximately 0.27 by the end of the reported epoch-40 update. The caption generation temperature is annealed from 1.0 to 0.8 with a cosine schedule over epochs 1–100, giving a temperature of approximately 0.933 at epoch 40.

B.4 Computational Resources

All reported training runs were launched as single-node distributed jobs using PyTorch DistributedDataParallel with `torchrun -nproc_per_node=3`. Each run used the three available NVIDIA A100-SXM4-80GB GPUs on the local machine; the display GPU was not used for training. The per-GPU batch size was 256 for MH communication, VLM updates, and ProbVLM updates, giving an effective batch size of 768 in the three-GPU distributed setting. For the main experiments, each encoder condition was run for 40 MHCG epochs with three seeds (seeds 1, 2, and 4; see Appendix B.5) unless otherwise noted. The HOMO, HETERO1, HETERO2, NoCom, and acceptance-rule ablation runs were launched sequentially by configuration and seed. Evaluation jobs for caption generation, retrieval, RSA, and ΔR^2 were run separately after training, typically using one GPU per evaluation process. Runtime estimates are based on W&B `_runtime` logs for completed 40-epoch COCO runs. Because the experiments were conducted on a shared machine where other processes could use the same GPUs concurrently, these values should be interpreted as observed wall-clock times rather than controlled throughput benchmarks. Across the completed main MHCG runs for HOMO, HETERO1, and HETERO2, a single 40-epoch run took 9.7–23.6 hours on the three-GPU setup, with a median of approximately 11.0 hours. This corresponds to a nominal 29–71 A100 GPU-hours per run.

B.5 Seed selection.

Seed 3 is excluded from all reported results because the ALLACCEPT condition under this initialization collapsed to a single repeated token sequence (token index 96 emitted at every position), making cross-condition comparisons within this seed unreliable. For reference, the MHCG condition under HETERO1 for seed 3 at epoch 40 yields the following retrieval and vision–text RSA values, which are consistent with, and in most directions slightly above, the seed-mean values reported in Table 3. Top-1 i2t retrieval (%, K=10): $A{\to}A = 93.1$, $A{\to}B = 90.4$, $B{\to}A = 90.3$, $B{\to}B = 91.6$. Top-1 t2i retrieval (%, K=10): $A{\to}A = 76.2$, $A{\to}B = 68.8$, $B{\to}A = 73.8$, $B{\to}B = 69.2$. Vision–text RSA: $\rho_s(\boldsymbol{t}^A, \boldsymbol{v}^A) = 0.153$, $\rho_s(\boldsymbol{t}^A, \boldsymbol{v}^B) = 0.149$, $\rho_s(\boldsymbol{t}^B, \boldsymbol{v}^A) = 0.150$, $\rho_s(\boldsymbol{t}^B, \boldsymbol{v}^B) = 0.154$. The seed exclusion is therefore conservative with respect to MHCG performance.

B.6 Evaluation Metric Details

This appendix gives the implementation-level definitions of the metrics summarized in Section 4. Retrieval accuracy, representational similarity, and statistics of the image sets associated with shared token sequences are measured on `val2017`. For visual-feature prediction, linear probes are trained on `train2017` and evaluated on `val2017`. Throughout, representational similarity analysis (RSA; Kriegeskorte et al. 12) compares representational dissimilarity matrices (RDMs) using Spearman correlations.

We use $X{\to}Y$ to denote a token-sequence-source/evaluation-space pairing. Agent X supplies the token sequence, and Agent Y supplies the representation space used for evaluation. This notation is independent of the retrieval mode and does not denote a communication or parameter-update pass.

Visual specificity of shared token sequences. To quantify how narrowly or broadly each shared token sequence groups images, we first identify shared token sequences within each condition and seed. For a token sequence s, the images jointly assigned by both agents to s are

$$\mathcal{I}_s = \{i : c_i^A = s \text{ and } c_i^B = s\},$$

where c_i^A and c_i^B are the greedy validation token sequences produced by Agents A and B for image i. Thus, $\mathcal{I}_s$ contains the images jointly assigned by both agents to the identical token sequence s. We retain token sequences with $|\mathcal{I}_s| \geq 10$ and call $|\mathcal{I}_s|$ the number of shared images for sequence s. The number of retained shared token sequences measures how many shared meanings are available in a condition. For each shared token sequence, we compute its global-normalized visual radius in three fixed CLS measurement spaces derived from frozen encoders: DINOv2 ViT-B, DINOv2 ViT-S, and MAE ViT-B. Let r_s^Z be the mean cosine distance from images in $\mathcal{I}_s$ to their centroid in measurement space Z, and let r_{global}^Z be the same centroid radius over all validation images in that measurement space. The global-normalized visual radius in measurement space Z is

$$r_s^Z / r_{\text{global}}^Z.$$

Lower values mean that the token sequence selects a tighter visual region than the validation set as a whole in that fixed measurement space. Within each seed, we summarize the distribution of global-normalized visual radii over shared token sequences by its mean and median; Table 4 then reports seed means and standard deviations of these summaries. The median shared images per sequence in Table 9 is the median of $|\mathcal{I}_s|$ across shared token sequences within a seed, again summarized across seeds. At the object-category level, we use COCO object annotations attached to the validation images in $\mathcal{I}_s$. Because a COCO image can contain multiple object categories, each present category contributes to the category-presence distribution. We compute its entropy $H(p_s)$ and report the effective number of object categories, $\exp(H(p_s))$; lower values indicate narrower object-level meanings. We also report the top-object mass, the largest category-presence probability in $\mathcal{I}_s$, where higher values indicate that a token sequence is more dominated by a single object category. All statistics in this paragraph are computed separately for each seed and condition before summarizing across seeds.

Visual-feature prediction. For each pairing $X \to Y$, we fit one-hot linear probes from Agent X's token indicators to Agent Y's fixed visual CLS principal-component scores. The probes are trained on `train2017` and evaluated on `val2017`. PCA is fit on Agent Y's `train2017` CLS features.

For each retained PC j, $R_j^2(X \to Y)$ denotes the validation R^2 of the probe trained with the true image–token pairing, and $R_{j,\text{perm}}^2(X \to Y)$ denotes the corresponding validation R^2 after shuffling train-set token rows within the token-sequence-source agent X. We compute these values for the top 64 PCs and report the explained-variance-weighted, permutation-corrected score:

$$\Delta R_{\text{vw}}^2(X \to Y) = \sum_{j=1}^{64} w_j^Y \big(R_j^2(X \to Y) - R_{j,\text{perm}}^2(X \to Y) \big), \tag{16}$$

where w_j^Y is the explained-variance ratio of PC j for Agent Y's visual features, normalized over the retained PCs. The permutation baseline is computed by repeating the train-set shuffling procedure five times; reported values average over the five permutations.

To characterize positional structure, we repeat the visual-feature prediction analysis for singleton token positions and for the full sequence. For singleton analyses, the one-hot predictors are built from a single token position rather than the full token sequence. Appendix Table 14 reports the cross-agent singleton-position scores for $A \to B$ and $B \to A$.

Representational Similarity Analysis (RSA). Inter-agent text–text RSA is the Spearman correlation ρ_s between the two agents' text RDMs, computed from normalized Hamming distances between token sequences. The Hamming distance is the fraction of mismatched token positions, with length differences counted as mismatches. We also track the number of unique token sequences to verify that these communication scores are not explained by token-sequence collapse.

For each pairing $X \to Y$, vision–text RSA compares Agent Y's visual RDM with Agent X's text RDM. The token-sequence RDM is the normalized Hamming-distance RDM defined above, and the visual RDM is the frozen-backbone CLS cosine-distance RDM computed with the deterministic validation transform. The score is

$$\rho_s \big(\text{RDM}_{\text{text}}^X, \text{RDM}_{\text{vis}}^Y \big) . \tag{17}$$

Vision–vision RSA, $\rho_s(\boldsymbol{v}^A, \boldsymbol{v}^B)$, is the Spearman correlation between the two agents' frozen visual CLS RDMs computed on the same validation images. The within-agent pairings ($A \to A$, $B \to B$)

measure how much visual information each agent's own token sequence carries. The cross-agent pairings ($A{\to}B$, $B{\to}A$) measure inter-agent transfer. To characterize positional structure, we repeat the visual-feature prediction analysis for singleton token positions and for the full sequence. Appendix C.6 reports the singleton-position analysis.

We also compute partial correlations to control the effect of visual similarity on vision–text RSA. Let $\rho_{X,Y|Z} = \rho_s(\mathrm{RDM}^X_{\text{text}}, \mathrm{RDM}^Y_{\text{vis}} \mid \mathrm{RDM}^Z_{\text{vis}})$ denote the Spearman partial correlation between Agent X's token-sequence RDM and Agent Y's visual RDM, controlling for Agent Z's visual RDM. Operationally, the RDM vectors entering each partial correlation are first converted to average ranks (ties receive their mean rank), centered, and then combined with the standard partial-correlation formula. This isolates the association between the token sequence and one agent's visual structure that is not explained by the other agent's visual structure. We summarize these as a *private-structure bias*:

$$\Delta_X = \rho_{X,X|Y} - \rho_{X,Y|X}, \tag{18}$$

where $\Delta_X \approx 0$ indicates that Agent X's token sequence reflects structure common to both visual spaces, and $\Delta_X > 0$ indicates that the token sequence disproportionately reflects Agent X's private visual structure. Results are reported in Appendix C.5.

Cross-agent retrieval. For a pairing $X{\to}Y$, Agent X's validation token sequence is encoded by Agent Y's text encoder, and the corresponding validation image is encoded by Agent Y's visual encoder. The projected image and text embeddings are ℓ_2-normalized and scored by dot product.

For each fixed pairing, we evaluate two retrieval modes. Image-to-text retrieval (i2t) uses an image embedding as the query and retrieves the matching token-sequence embedding. Text-to-image retrieval (t2i) uses a token-sequence embedding as the query and retrieves the matching image embedding. Here, "text" refers to the emergent token sequence rather than natural language.

For each query, the candidate set contains the true paired item and $K-1$ distractors sampled without replacement. We average over candidate-sampling seeds 200–299 and evaluate all validation queries unless otherwise stated. Appendix Table 8 reports top-1 accuracy with $K{=}10$ for all four pairings ($A{\to}A$, $A{\to}B$, $B{\to}A$, $B{\to}B$), with i2t and t2i shown separately.

Positional entropy. To measure how actively each token position is used, we compute the entropy of the empirical token distribution at each position, separately for each agent, seed, condition, and method. If $p_{a,\ell}(v)$ is the validation-set frequency with which Agent a emits token v at position ℓ, the positional entropy is

$$H_{a,\ell} = -\sum_v p_{a,\ell}(v) \log_2 p_{a,\ell}(v).$$

Higher entropy indicates more diverse use of that position, while near-zero entropy indicates an effectively fixed or inactive position. Appendix Table 13 reports these values in bits.

C Details of Main Results

This section expands the main-text summary results with the seed-averaged statistics, direction-wise scores, and qualitative examples used to interpret them. The subsections follow the same order as the main Results section: full communication performance, training dynamics, visual specificity of shared token sequences, qualitative boundary shifts, partial-RSA asymmetry, and position-wise analyses.

C.1 Full direction-wise communication results

The main summary table averages over communication directions to show the main trend. Table 8 keeps the full direction-wise structure: within-agent directions ($A{\to}A$ and $B{\to}B$) indicate how much visual information a token sequence carries about its own agent's visual space, whereas cross-agent directions ($A{\to}B$ and $B{\to}A$) test whether that information transfers to the other agent's space.

Across conditions, MHCG improves the cross-agent directions over NoCom in visual-feature prediction, RSA, and retrieval. The difference between within-agent and cross-agent scores becomes most informative under heterogeneity: within-agent scores can remain nontrivial even without successful

Table 8: Direction-wise communication accuracy and visual information results at epoch 40 on COCO `val2017`. Seed means ± SD over three seeds. For HETERO1, acceptance-rule ablations are also shown. Visual feature prediction reports the explained-variance-weighted real-minus-image-permutation ΔR^2 for token-to-vision linear prediction over the top 64 visual PCs. RSA reports Spearman correlations among text and vision RDMs; vision–vision RSA is fixed by the encoder pair and is therefore shown without a standard deviation. Retrieval reports top-1 i2t and t2i accuracy (K=10, chance = 10%) as percentages for all four directions ($X \to Y$: Agent X's text is evaluated in Agent Y's multimodal space).

(a) Visual feature prediction

Condition	Method	$\Delta R^2_{A \to A}$	$\Delta R^2_{A \to B}$	$\Delta R^2_{B \to A}$	$\Delta R^2_{B \to B}$
HOMO	MHCG	$\underline{0.310}\pm0.034$	$\underline{0.310}\pm0.034$	$\underline{0.313}\pm0.035$	$\underline{0.313}\pm0.035$
	NoCom	0.057 ± 0.015	0.057 ± 0.015	0.043 ± 0.018	0.043 ± 0.018
HETERO1	MHCG	$\underline{0.308}\pm0.001$	$\underline{0.271}\pm0.008$	$\underline{0.301}\pm0.006$	$\underline{0.273}\pm0.009$
	NoCom	0.057 ± 0.015	0.054 ± 0.014	0.024 ± 0.017	0.025 ± 0.019
	AllAccept	0.037 ± 0.036	0.036 ± 0.034	0.038 ± 0.035	0.038 ± 0.035
	Random accept	0.010 ± 0.005	0.010 ± 0.005	0.009 ± 0.008	0.006 ± 0.007
	ITM-based	0.099 ± 0.033	0.098 ± 0.028	0.100 ± 0.032	0.102 ± 0.028
HETERO2	MHCG	$\underline{0.162}\pm0.058$	$\underline{0.126}\pm0.088$	$\underline{0.111}\pm0.058$	0.127 ± 0.106
	NoCom	0.057 ± 0.015	0.037 ± 0.026	0.013 ± 0.006	$\underline{0.335}\pm0.252$

(b) Representational Similarity Analysis

Condition	Method	$\rho_s(\boldsymbol{t}^A, \boldsymbol{t}^B)$	$\rho_s(\boldsymbol{v}^A, \boldsymbol{v}^B)$	$\rho_s(\boldsymbol{t}^A, \boldsymbol{v}^A)$	$\rho_s(\boldsymbol{t}^A, \boldsymbol{v}^B)$	$\rho_s(\boldsymbol{t}^B, \boldsymbol{v}^A)$	$\rho_s(\boldsymbol{t}^B, \boldsymbol{v}^B)$
HOMO	MHCG	$\underline{0.937}\pm0.021$	1.000	$\underline{0.161}\pm0.054$	$\underline{0.161}\pm0.054$	$\underline{0.162}\pm0.054$	$\underline{0.162}\pm0.054$
	NoCom	0.044 ± 0.057		0.082 ± 0.027	0.082 ± 0.027	0.075 ± 0.015	0.075 ± 0.015
HETERO1	MHCG	$\underline{0.872}\pm0.032$	0.417	$\underline{0.139}\pm0.033$	$\underline{0.119}\pm0.051$	$\underline{0.134}\pm0.028$	$\underline{0.118}\pm0.048$
	NoCom	0.042 ± 0.012		0.082 ± 0.027	0.067 ± 0.018	0.039 ± 0.048	0.041 ± 0.041
	AllAccept	0.774 ± 0.236		0.109 ± 0.002	0.090 ± 0.050	0.098 ± 0.009	0.080 ± 0.036
	Random accept	0.011 ± 0.023		0.025 ± 0.013	0.021 ± 0.035	0.012 ± 0.025	0.008 ± 0.015
	ITM-based	0.718 ± 0.166		0.106 ± 0.038	0.094 ± 0.047	0.105 ± 0.041	0.094 ± 0.052
HETERO2	MHCG	$\underline{0.673}\pm0.086$	0.074	$\underline{0.134}\pm0.016$	$\underline{0.063}\pm0.056$	$\underline{0.112}\pm0.039$	0.078 ± 0.058
	NoCom	0.015 ± 0.005		0.082 ± 0.027	0.042 ± 0.030	0.032 ± 0.005	$\underline{0.298}\pm0.113$

(c) Cross-modal retrieval

Condition	Method	Top-1 i2t				Top-1 t2i			
		A→A	A→B	B→A	B→B	A→A	A→B	B→A	B→B
HOMO	MHCG	$\underline{92.4}\pm1.5$	$\underline{92.3}\pm1.1$	$\underline{92.3}\pm1.6$	$\underline{92.5}\pm1.3$	$\underline{75.6}\pm4.7$	$\underline{75.5}\pm4.3$	$\underline{75.4}\pm4.8$	$\underline{75.5}\pm4.5$
	NoCom	48.8 ± 7.8	23.4 ± 3.9	22.6 ± 10.7	54.9 ± 1.2	23.0 ± 4.3	9.8 ± 0.6	9.4 ± 0.8	21.4 ± 3.6
HETERO1	MHCG	$\underline{89.7}\pm2.8$	$\underline{87.6}\pm2.6$	$\underline{87.5}\pm2.6$	$\underline{88.7}\pm2.7$	$\underline{65.9}\pm7.8$	$\underline{59.6}\pm6.6$	$\underline{64.4}\pm7.6$	$\underline{60.2}\pm6.8$
	NoCom	48.8 ± 7.8	19.8 ± 2.6	34.3 ± 9.5	40.4 ± 10.3	23.0 ± 4.3	9.5 ± 0.8	10.1 ± 0.3	12.7 ± 1.9
	AllAccept	52.6 ± 3.1	56.2 ± 0.4	49.5 ± 6.5	52.3 ± 5.5	16.2 ± 5.4	15.5 ± 4.9	16.1 ± 5.3	15.5 ± 4.9
	Random accept	33.3 ± 14.8	43.6 ± 9.7	52.0 ± 4.3	50.9 ± 5.4	10.0 ± 0.0	10.1 ± 0.0	10.0 ± 0.0	10.1 ± 0.0
	ITM-based	70.3 ± 9.3	66.9 ± 9.9	68.6 ± 9.6	68.4 ± 10.0	41.2 ± 8.7	35.6 ± 9.5	40.9 ± 8.4	36.5 ± 9.1
HETERO2	MHCG	$\underline{78.3}\pm7.4$	$\underline{47.7}\pm13.8$	$\underline{70.3}\pm3.7$	$\underline{53.4}\pm12.1$	$\underline{41.6}\pm8.0$	$\underline{37.8}\pm12.1$	$\underline{33.2}\pm12.5$	$\underline{36.2}\pm13.1$
	NoCom	48.8 ± 7.8	20.1 ± 7.4	24.7 ± 13.3	32.2 ± 1.5	23.0 ± 4.3	9.9 ± 0.8	10.2 ± 0.4	13.6 ± 4.9

coordination, but the cross-agent directions reveal whether the token sequence carries visual structure that the other agent can use. Among the HETERO1 acceptance controls, MHCG is the only method that is consistently strongest across visual prediction, vision–text RSA, and both retrieval modes.

C.2 Training dynamics for communication results

The epoch-40 tables summarize the endpoint of learning, but they do not show whether the final system emerges through stable growth or through transient collapse. Figure 5 therefore plots retrieval, inter-agent token-sequence RSA, token-sequence diversity, and MH acceptance over training for the three encoder conditions.

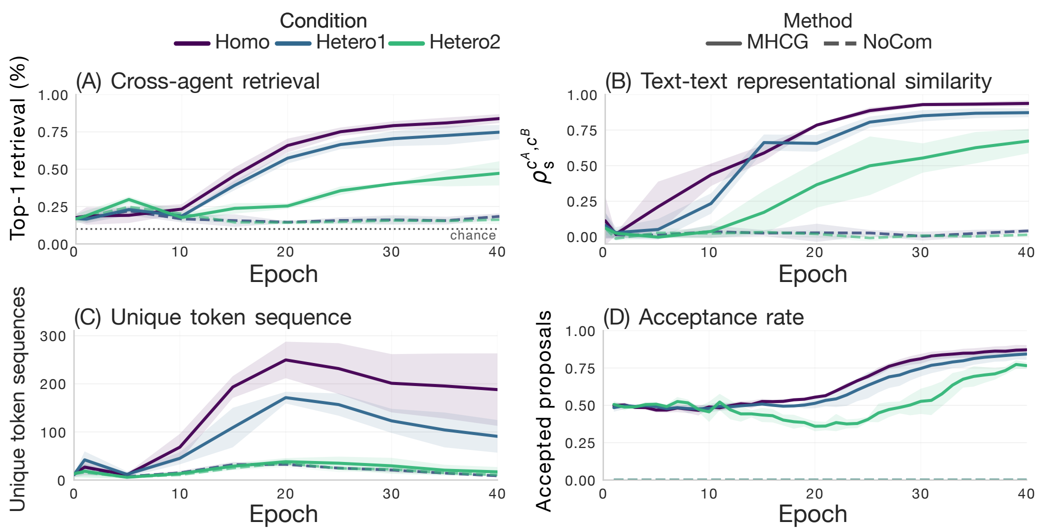


Figure 5: Token-sequence diversity, training dynamics, and token-sequence alignment over training across the three encoder conditions, each compared against its NoCom baseline. Colors indicate encoder condition (HOMO, HETERO1, HETERO2); solid curves show MHCG and dashed curves show NoCom. Shaded bands indicate mean ± standard deviation across available seeds. (A) Cross-agent top-1 retrieval, averaged over the token-sequence-source/evaluation-space pairings $A \to B$ and $B \to A$ and over image-to-text (i2t) and text-to-image (t2i) retrieval modes ($K = 10$, chance $= 10\%$); (B) inter-agent text–text RSA. (C) unique token sequences, averaged over agents; (D) mean acceptance rate, averaged over directions.

Figure 5 shows the learning dynamics, associated token-sequence diversity, and MH acceptance traces corresponding to the epoch-40 retrieval values in Appendix Table 8. MHCG increases cross-agent retrieval and inter-agent token-sequence alignment while maintaining a non-collapsed number of unique token sequences. The HETERO2 curves remain lower and less stable, matching the coarser endpoint statistics in the main and appendix tables. Figure 6 provides the analogous training dynamics for the HETERO1 acceptance-rule controls.

The ablation dynamics clarify why listener-side MH acceptance matters. AllAccept can produce high inter-agent token-sequence RSA, but this is paired with sharply reduced token-sequence diversity and weak retrieval, consistent with degenerate agreement rather than useful communication. Random accept shows that a similar acceptance rate is not sufficient by itself; the acceptance decision must be tied to the listener's perceptual evidence. ITM-based accept is intermediate, whereas MHCG is the only setting that jointly preserves diversity and high cross-agent retrieval.

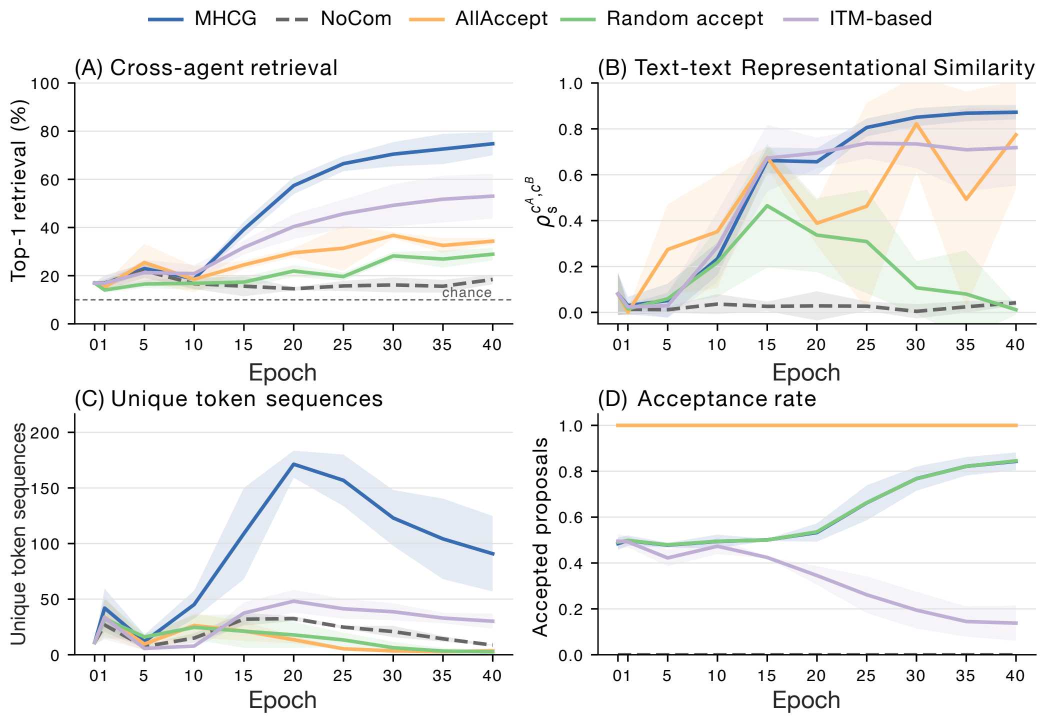


Figure 6: HETERO1 acceptance-rule ablation dynamics. Curves compare MHCG, NoCom, AllAccept, Random accept, and ITM-based accept over training. Shaded bands indicate mean $\pm$ SD across three seeds. (A) Cross-agent top-1 retrieval, averaged over $A \to B$ and $B \to A$ and over i2t/t2i retrieval modes; (B) inter-agent text–text RSA based on normalized Hamming RDMs; (C) unique token sequences, averaged over agents; (D) mean acceptance rate, averaged over directions.

C.3 Visual specificity of shared token sequences

Table 9 complements the fixed-space visual-radius analysis in Table 4 by asking how broad each shared token sequence is in terms of images and COCO object categories. The shared-sequence count captures how many shared token sequences survive the support threshold, whereas shared images per sequence, effective object categories, and top-object mass characterize how broad those shared meanings are.

Table 9: Object-level breadth of shared token sequences. A token sequence is shared within a seed if at least 10 validation images receive that sequence from both agents. The remaining quantities are medians over shared token sequences within a seed, then averaged across three seeds. Fewer shared images per sequence and fewer effective object categories indicate narrower shared meanings; higher top-object mass indicates stronger concentration on a dominant object category.

Condition	Shared token sequences ↑	Shared images per sequence ↓	Effective object categories ↓	Top-object mass ↑
HOMO	77.7±18.5	32.7±6.7	9.5±1.0	0.942±0.050
HETERO1	47.3±9.3	47.3±4.2	8.7±1.6	0.985±0.014
HETERO2	11.0±7.2	547.7±761.8	20.7±9.9	0.834±0.224

HETERO1 forms fewer shared token sequences than HOMO, but those sequences are more object-concentrated: they have lower effective object-category breadth and higher top-object mass. Because this smaller sequence count could in principle bias a direct radius comparison, Table 10 provides a matched-size bootstrap control in which HOMO is subsampled to HETERO1's token count; the matched HOMO radii are unchanged, confirming that the per-sequence concentration difference is not explained by HOMO having a long tail of highly concentrated sequences. The higher object concentration of HETERO1 sequences is therefore consistent with the selection-bottleneck interpretation in the main text: heterogeneous constraints retain only encoder-invariant visual concepts, resulting in fewer but more concentrated shared sequences. The reduced sequence count contributes to lower system-level vocabulary coverage, consistent with the lower ΔR^2 under HETERO1. In contrast, HETERO2 yields very few shared token sequences with much more shared images per sequence and broader object coverage, consistent with coarse token meanings under stronger visual mismatch.

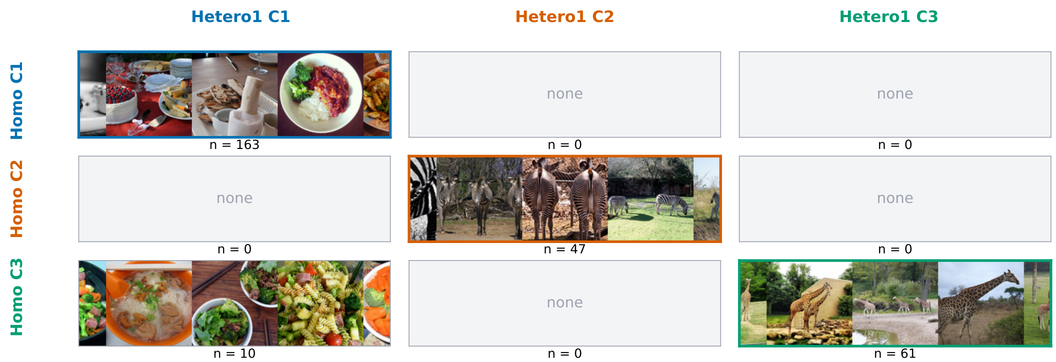
Table 10: Matched-size bootstrap control for shared-sequence count. For each seed, HOMO is subsampled to $k = \min(n_{\text{Homo}}, n_{\text{Hetero1}})$ tokens without replacement (2,000 iterations); HETERO1 uses all k tokens. The metric is the global-normalised visual radius r/r_{global} in the DINOv2 ViT-B CLS space (lower = more visually specific). The rightmost column shows the 95% bootstrap CI of the per-iteration difference ($\text{HOMO} - \text{HETERO1}$).

Seed	k	Condition	Original mean	Matched mean	95% CI of diff
1	58	HOMO	0.539	0.539	$[-0.037,\ +0.033]$
		HETERO1	0.542	0.542	
2	43	HOMO	0.508	0.508	$[-0.049,\ +0.032]$
		HETERO1	0.517	0.517	
4	41	HOMO	0.527	0.527	$[+0.033,\ +0.126]$
		HETERO1	0.447	0.447	

C.4 Additional Qualitative Examples and Boundary Shifts

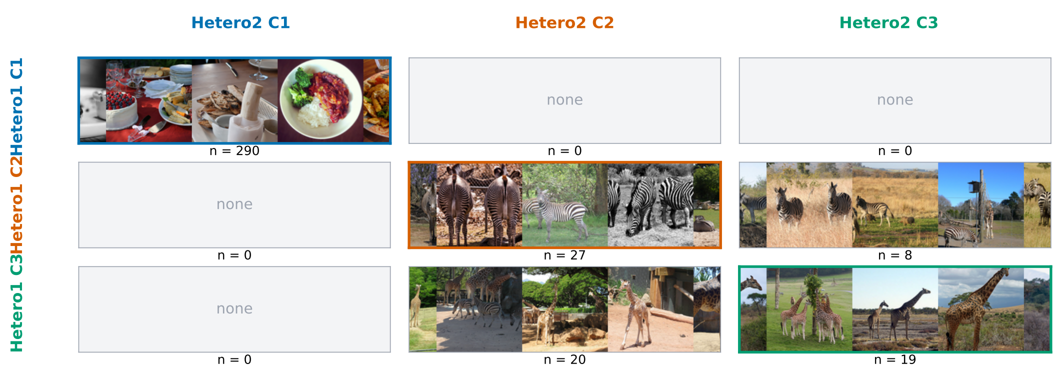
Figure 2 shows representative nearest-neighbor examples for token-sequence concepts matched across encoder conditions by image-set overlap. We also compare the matched HOMO and HETERO1 C1–C3 sets of images jointly assigned by both agents to the corresponding token sequence. Figure 7 shows the full cross-assignment matrices for all three condition pairs. For HOMO vs. HETERO1, most matched concepts overlap on the diagonal, with the only nonzero off-diagonal cell being HOMO C3 intersected with HETERO1 C1 (food-like scenes). The HOMO vs. HETERO2 and HETERO1 vs. HETERO2 matrices show the corresponding overlap structure under stronger encoder mismatch.

Here we provide complementary examples for the ten most frequent shared token patterns within each encoder condition. For each pattern, nearest neighbors are selected around the pattern centroid in each agent's frozen CLS visual space. When available, the centroid is computed from images for which both agents assign that pattern to the same image; otherwise, the candidate set falls back to images for which either agent produces that pattern.



(a) HOMO vs. HETERO1

(b) HOMO vs. HETERO2



(c) HETERO1 vs. HETERO2

Figure 7: C1–C3 cross-assignment matrices for all condition pairs. Each cell shows representative images jointly assigned by both agents to the corresponding token-sequence concept row and column. Concepts are the matched C1–C3 image-set concepts used in Figure 2; images are shown in the fixed DINOv2 ViT-B Agent-A measurement space.

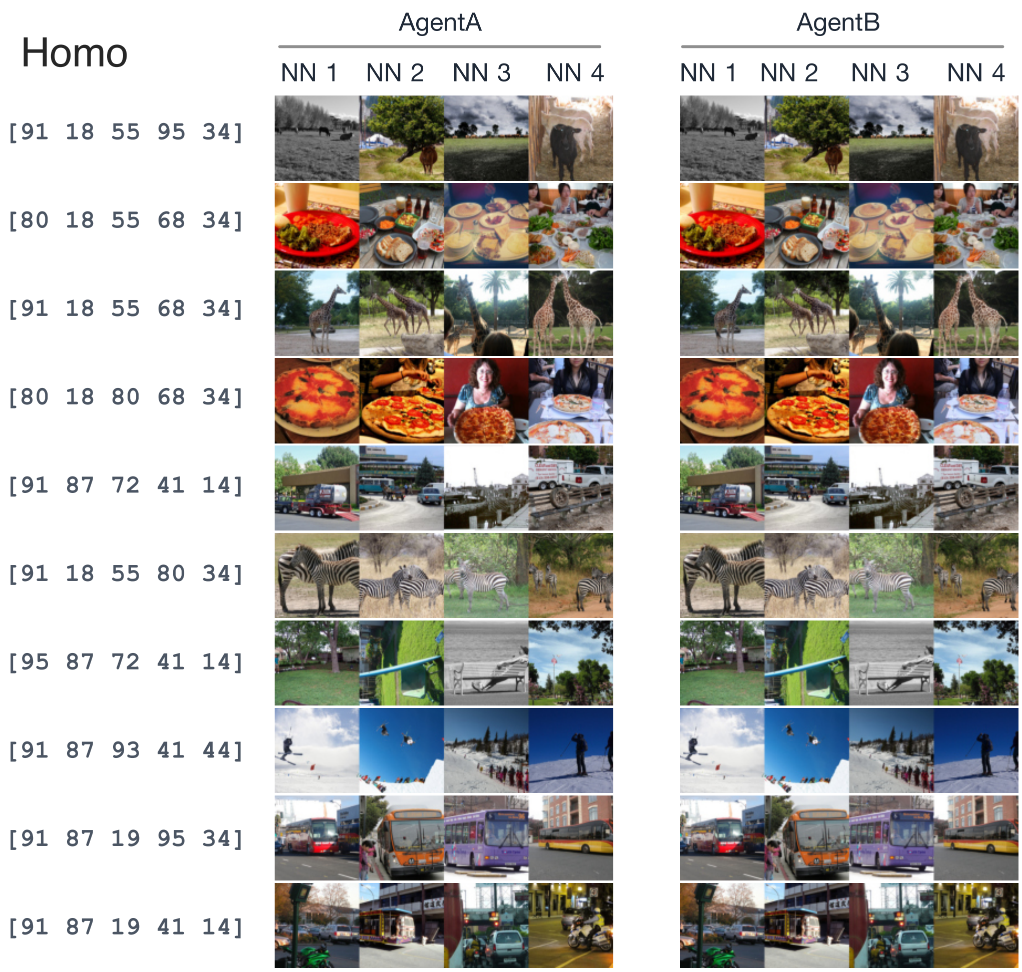


Figure 8: Additional nearest-neighbor examples for HOMO. Rows are the ten most frequent shared token patterns. The left and right image strips show nearest neighbors in Agent A's and Agent B's frozen CLS spaces, respectively.

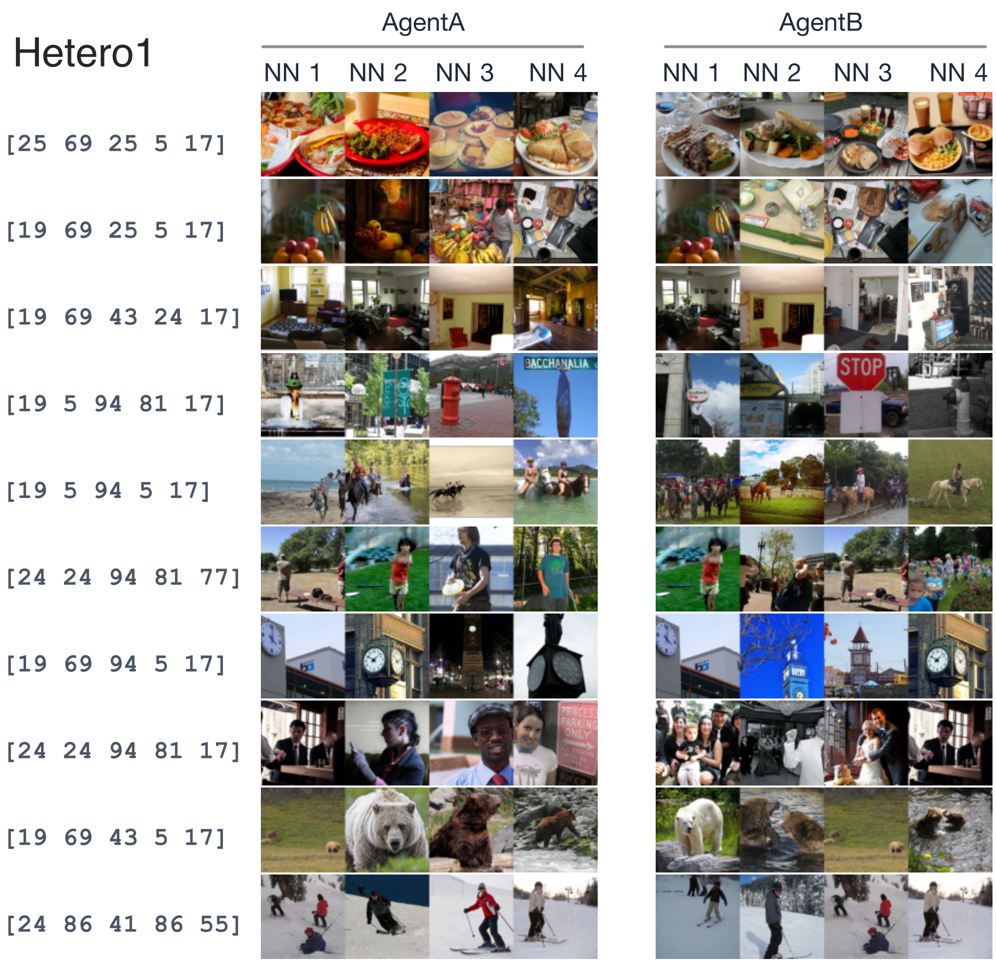


Figure 9: Additional nearest-neighbor examples for HETERO1. Rows are the ten most frequent shared token patterns. The left and right image strips show nearest neighbors in Agent A's and Agent B's frozen CLS spaces, respectively.

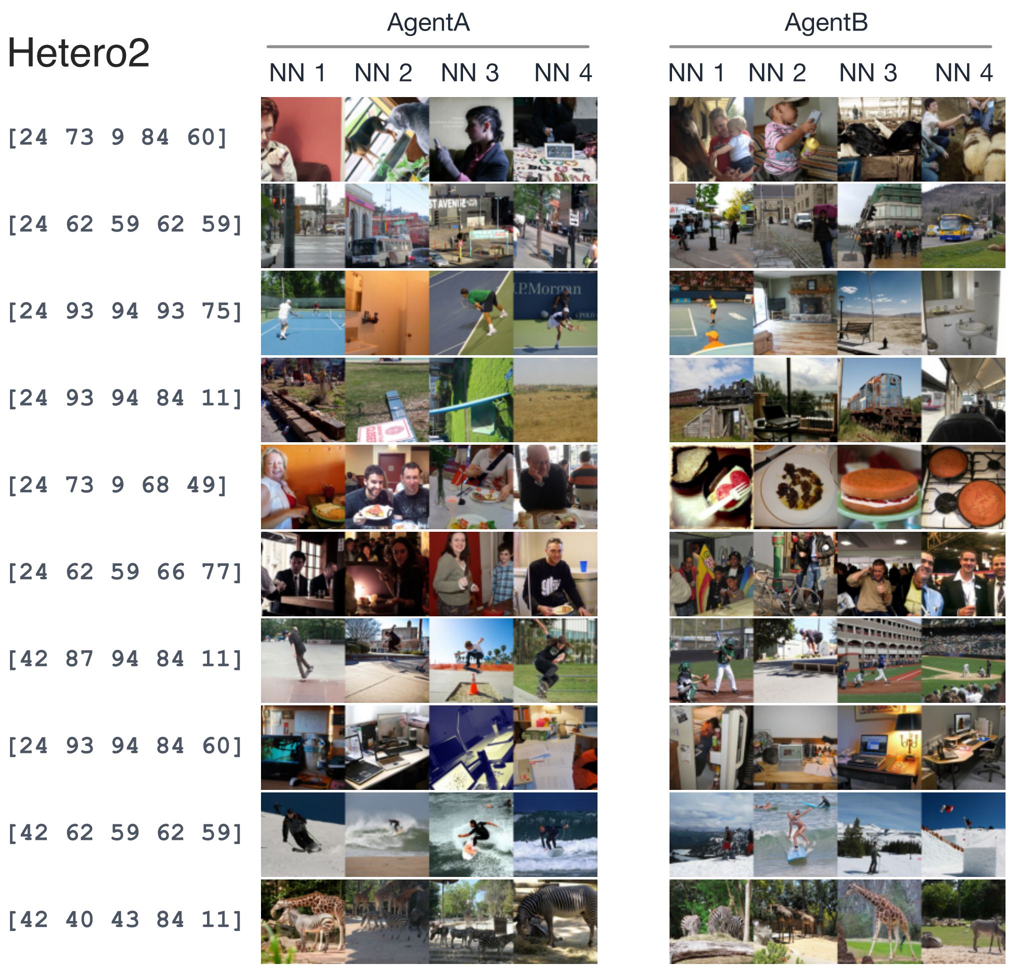


Figure 10: Additional nearest-neighbor examples for HETERO2. Rows are the ten most frequent shared token patterns. The left and right image strips show nearest neighbors in Agent A's and Agent B's frozen CLS spaces, respectively.

C.5 Partial Correlation Summary

Appendix Table 8 reports vision–text RSA in four directions. Comparing within-agent values $(\boldsymbol{t}^X, \boldsymbol{v}^X)$ with cross-agent values $(\boldsymbol{t}^X, \boldsymbol{v}^Y)$ provides an initial indication of whether the token sequence reflects structure common to both agents or private to one. Partial correlations provide a more formal decomposition by asking whether a token-sequence RDM remains related to one visual RDM after controlling for the other visual RDM.

Table 11 reports partial Spearman correlations at epoch 40. Because HOMO uses identical visual RDMs, its partial correlations are undefined and omitted from the table. In HETERO1, Δ_A and Δ_B are small, consistent with the relatively symmetric pattern observed in the main-text ΔR^2 and RSA values. Under HETERO2, $\Delta_A > \Delta_B$, confirming that Agent A's (DINOv2) token sequence disproportionately reflects its own visual structure. These partial-correlation results are consistent with the within-agent vs. cross-agent comparisons in Table 8 and provide a complementary statistical perspective.

Table 11: Partial Vision–Text RSA at epoch 40 on COCO `val2017`. Means ± SD over three seeds. Each entry is $\rho_{X,Y|Z} = \rho_s(\mathrm{RDM}^X_{\text{text}}, \mathrm{RDM}^Y_{\text{vis}} \mid \mathrm{RDM}^Z_{\text{vis}})$, partial correlation between Agent X's Hamming-distance token-sequence RDM and Agent Y's visual RDM, controlling for Agent Z's visual RDM. The final columns report $\Delta_A = \rho_{A,A|B} - \rho_{A,B|A}$ and $\Delta_B = \rho_{B,B|A} - \rho_{B,A|B}$, computed per seed before averaging. HOMO is omitted because the two visual RDMs are identical, making the partial correlations undefined.

Condition	Method	Partial ρ_s				Bias	
		$A,A\|B$	$A,B\|A$	$B,A\|B$	$B,B\|A$	Δ_A	Δ_B
HETERO1	MHCG	0.099±0.014	0.068±0.041	0.093±0.010	0.070±0.040	0.031±0.029	−0.024±0.031
	NoCom	0.059±0.022	0.037±0.008	0.024±0.035	0.028±0.023	0.022±0.016	0.004±0.013
HETERO2	MHCG	0.130±0.013	0.053±0.056	0.107±0.035	0.070±0.056	0.077±0.050	−0.037±0.021
	NoCom	0.079±0.026	0.036±0.028	0.010±0.011	0.297±0.114	0.043±0.010	0.287±0.124

The normalized Hamming RDM contains many tied entries, with tie density ranging from 0.19 (HETERO1 MHCG) to 0.66 (HETERO1 NoCom seed 2). Spearman partial correlations on heavily tied RDMs may be sensitive to the mid-rank convention. We therefore report a Kendall τ_b-based partial-RSA analogue as a tie-aware robustness check, computed by applying the standard partial-correlation algebra $\tau_{X,Y|Z} = (\tau_{XY} - \tau_{XZ}\tau_{YZ})/\sqrt{(1-\tau^2_{XZ})(1-\tau^2_{YZ})}$, where each τ is the corresponding pairwise Kendall τ_b value.

Table 12 compares Spearman and Kendall τ_b partial-RSA bias values. The two metrics differ in absolute magnitude (Kendall values are approximately 0.6–0.8 times the corresponding Spearman values, consistent with the standard relationship between the two correlation measures), but agree in sign and in the qualitative pattern across conditions. The MHCG-versus-NoCom shift in Δ_B under HETERO2 is preserved. The Δ_B sign reversal between NoCom and MHCG is observed in all three seeds under both metrics.

Table 12: Partial-RSA under Kendall τ_b. Spearman partial correlations are reproduced from Table 11; Kendall τ_b-based partial scores are computed by applying the partial-correlation formula to pairwise Kendall τ_b values.

		Spearman partial		Kendall τ_b-based partial	
Condition	Method	Δ_A	Δ_B	Δ_A	Δ_B
HETERO1	MHCG	+0.031±0.029	−0.024±0.031	+0.020±0.018	−0.015±0.020
	NoCom	+0.022±0.016	+0.004±0.013	+0.014±0.010	+0.003±0.009
HETERO2	MHCG	+0.077±0.050	−0.037±0.021	+0.056±0.036	−0.028±0.017
	NoCom	+0.043±0.010	+0.287±0.124	+0.031±0.008	+0.212±0.090

C.6 Positional Analyses

The previous analyses treat each length-5 token sequence as a single symbolic unit. The positional analyses ask whether the communication system distributes information across positions or relies on a small subset of positions. Positional entropy measures how actively each position varies, and singleton-position ΔR^2 measures how much cross-agent visual information can be decoded from one position at a time.

Table 13: Positional entropy of generated token sequences. Mean $\pm$ SD in bits across three seeds at epoch 40.

Condition	Method	Agent	Pos. 1	Pos. 2	Pos. 3	Pos. 4	Pos. 5
HOMO	MHCG	A	2.25 ± 0.30	2.15 ± 0.60	1.99 ± 0.33	1.78 ± 0.52	2.01 ± 0.60
		B	2.25 ± 0.29	2.18 ± 0.57	2.00 ± 0.32	1.79 ± 0.54	2.01 ± 0.57
	NoCom	A	1.21 ± 0.30	1.26 ± 0.64	0.97 ± 0.51	0.95 ± 0.12	0.82 ± 0.72
		B	0.91 ± 0.39	0.85 ± 0.16	0.76 ± 0.78	1.21 ± 0.35	0.64 ± 0.56
HETERO1	MHCG	A	1.71 ± 0.22	2.56 ± 0.11	1.95 ± 0.46	2.04 ± 0.71	1.51 ± 0.26
		B	1.72 ± 0.25	2.56 ± 0.11	1.97 ± 0.46	2.02 ± 0.71	1.52 ± 0.24
	NoCom	A	1.21 ± 0.30	1.26 ± 0.64	0.97 ± 0.51	0.95 ± 0.12	0.82 ± 0.72
		B	0.09 ± 0.07	0.30 ± 0.30	0.07 ± 0.06	0.36 ± 0.56	0.70 ± 0.45
HETERO2	MHCG	A	0.21 ± 0.29	1.33 ± 0.95	1.69 ± 0.52	1.67 ± 0.50	1.62 ± 0.78
		B	0.21 ± 0.35	1.28 ± 1.02	1.58 ± 0.54	1.60 ± 0.54	1.27 ± 1.16
	NoCom	A	1.21 ± 0.30	1.26 ± 0.64	0.97 ± 0.51	0.95 ± 0.12	0.82 ± 0.72
		B	0.49 ± 0.39	0.54 ± 0.52	0.97 ± 0.15	0.89 ± 0.68	1.00 ± 0.46

Table 14: Cross-agent singleton-position visual-feature prediction. Values are explained-variance-weighted ΔR^2 for singleton token positions at epoch 40, reported as mean $\pm$ standard deviation across three seeds. The $A \to B$ direction predicts Agent B's visual PCs from Agent A's singleton token position, and $B \to A$ is defined analogously.

Condition	Method	Direction	Pos. 1	Pos. 2	Pos. 3	Pos. 4	Pos. 5
HOMO	MHCG	$A \to B$	0.111 ± 0.028	0.084 ± 0.028	0.097 ± 0.024	0.099 ± 0.029	0.096 ± 0.021
		$B \to A$	0.110 ± 0.024	0.085 ± 0.026	0.099 ± 0.024	0.100 ± 0.030	0.096 ± 0.021
	NoCom	$A \to B$	0.029 ± 0.020	0.030 ± 0.015	0.016 ± 0.002	0.010 ± 0.006	0.015 ± 0.013
		$B \to A$	0.021 ± 0.009	0.019 ± 0.003	0.018 ± 0.018	0.023 ± 0.007	0.011 ± 0.010
HETERO1	MHCG	$A \to B$	0.080 ± 0.036	0.130 ± 0.015	0.094 ± 0.028	0.082 ± 0.035	0.071 ± 0.011
		$B \to A$	0.092 ± 0.048	0.141 ± 0.027	0.107 ± 0.031	0.085 ± 0.030	0.078 ± 0.017
	NoCom	$A \to B$	0.028 ± 0.018	0.029 ± 0.015	0.016 ± 0.002	0.010 ± 0.005	0.014 ± 0.012
		$B \to A$	0.004 ± 0.003	0.008 ± 0.004	0.002 ± 0.003	0.009 ± 0.014	0.019 ± 0.021
HETERO2	MHCG	$A \to B$	0.008 ± 0.012	0.081 ± 0.080	0.084 ± 0.076	0.068 ± 0.053	0.072 ± 0.043
		$B \to A$	0.011 ± 0.020	0.062 ± 0.033	0.059 ± 0.044	0.061 ± 0.048	0.056 ± 0.030
	NoCom	$A \to B$	0.019 ± 0.014	0.030 ± 0.027	0.015 ± 0.017	0.013 ± 0.020	0.017 ± 0.018
		$B \to A$	0.003 ± 0.001	0.003 ± 0.003	0.006 ± 0.004	0.005 ± 0.004	0.006 ± 0.001

Tables 13 and 14 report how actively each token position is used and how much cross-agent visual information each singleton position carries for MHCG and the NoCom baseline. Under HETERO1, both agents use all positions with comparable entropy, and singleton positions, especially positions 2–3, carry cross-agent visual information in both directions. Under HETERO2, positional entropy is more uneven and cross-agent singleton ΔR^2 is weaker and more directionally asymmetric. This compression of position-wise cross-agent visual information is consistent with the finding that, under strong heterogeneity, each shared token sequence tends to cover broader image sets.

D Broader Impacts

This work is foundational research on emergent communication and does not target a specific deployment. We identify the following potential societal impacts.

Positive impacts. Understanding how agents with heterogeneous private perceptual systems can develop shared communication through decentralized interaction informs the design of distributed multi-agent AI systems that must coordinate across diverse architectures without centralised control. The decentralized update rule, which requires no parameter sharing, cross-agent gradients, or shared reward signal, may contribute to privacy-preserving multi-agent learning, where agents coordinate without exposing their internal representations to one another. At a scientific level, this work contributes to understanding how shared symbolic systems can arise from diverse sensory experiences, with potential relevance to cognitive science and the study of language grounding.

Negative impacts and mitigations. The primary concern identified by our own results is *representational bias propagation*: emergent shared representations systematically inherit one agent's visual geometry rather than reflecting a neutral average of both. If systems of this kind were scaled or deployed, encoder-specific perceptual biases, including those arising from training data or architectural choices, could be embedded in a shared language space in ways that are difficult to detect or audit. The representational bias analysis introduced in this work (Section 5, Table 5) provides a diagnostic framework for quantifying such asymmetries, which could be applied to audit bias in deployed emergent communication systems. A second, more distal concern is that autonomous inter-agent communication protocols developed through decentralised interaction could, in more capable settings, become difficult for humans to interpret or monitor; the small vocabulary ($|\mathcal{V}| = 100$) and sequence length ($L = 5$) used here are far from such a regime. We do not foresee a direct path from the present work to harmful applications, but we consider transparency about these considerations appropriate.